

SECCL1

Identity and Access Management in SAP Cloud Platform

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

Course Version: 11
Course Duration: 3 Day(s)
e-book Duration: 1 Hour 45 Minutes
Material Number: 50146810

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	Example text
Window title	Example text

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Course Overview

TARGET AUDIENCE

This course is intended for the following audiences:

- Enterprise Architect
- Solution Architect
- System Administrator
- System Architect

UNIT 1

SAP Cloud Platform Security

Lesson 1

Describing SAP Cloud Platform and SAP Security Portfolio

2

Lesson 2

Describing the Security Setup in SAP Cloud Platform

6

UNIT OBJECTIVES

Describe SAP Cloud Platform and SAP Security Portfolio

Describe the security setup in SAP Cloud Platform

Unit 1

Lesson 1

Describing SAP Cloud Platform and SAP Security Portfolio

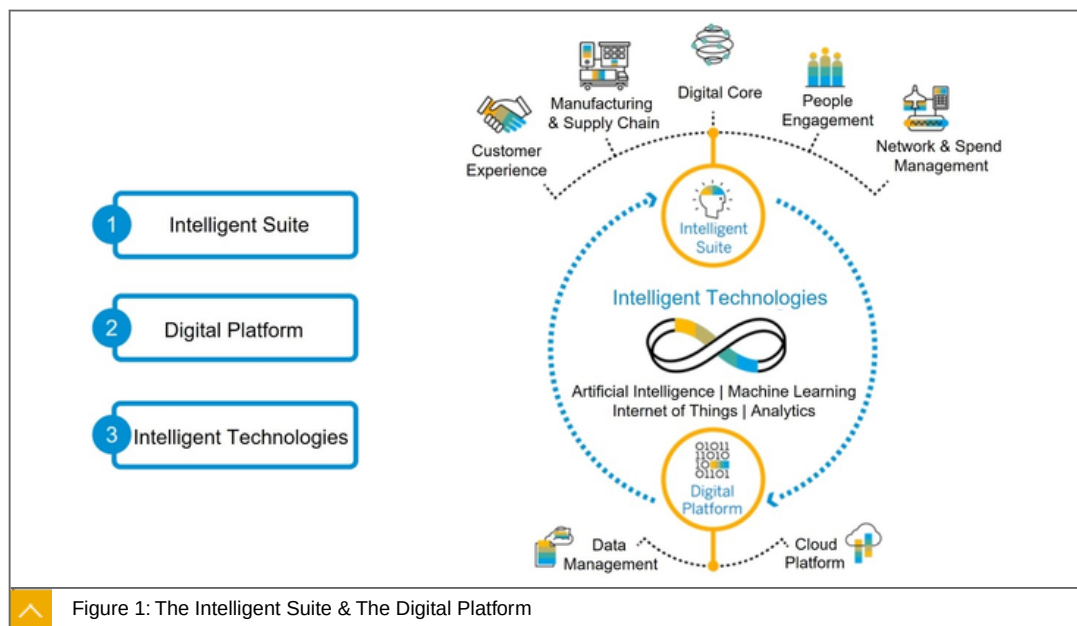


LESSON OBJECTIVES

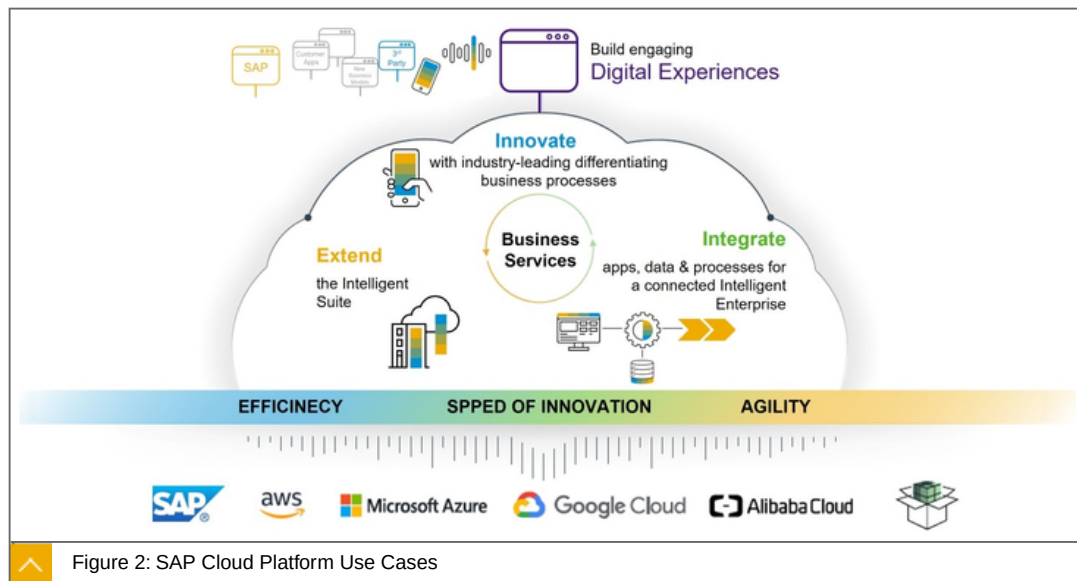
After completing this lesson, you will be able to:

Describe SAP Cloud Platform and SAP Security Portfolio

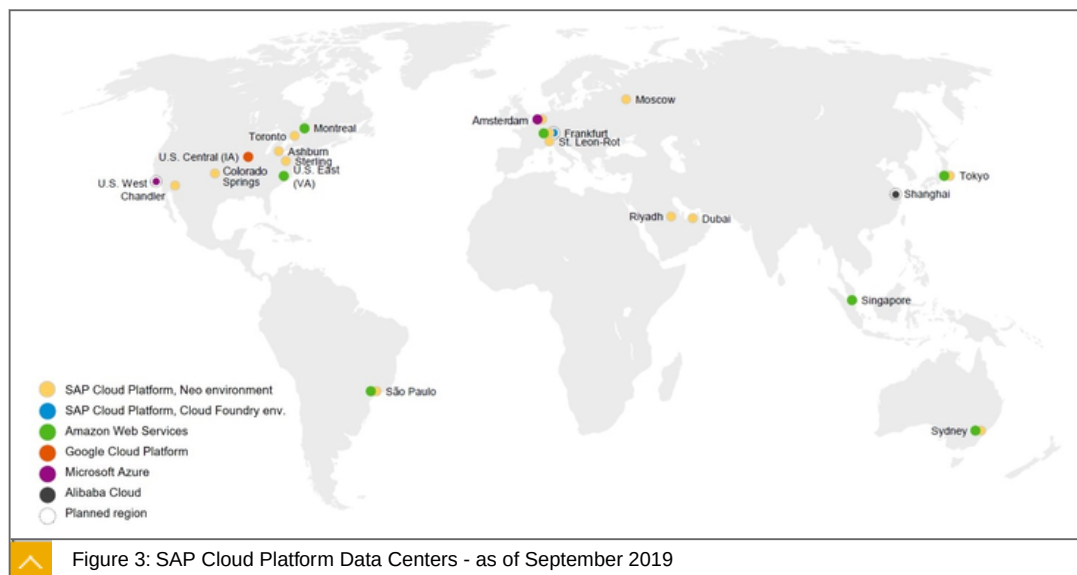
SAP Cloud Platform and SAP Security Portfolio



SAP future strategy is built on three main pillars: The Intelligent Suite, Digital Platform, and Intelligent Technologies. While all three pillars play an important part in delivering a first-class digital experience to customers, within this course we are going to focus on the Digital Platform and more specifically on the Cloud Platform subsection.



The Cloud Platform offers numerous use cases, which can be summarized as: Innovate, Extend, and Integrate. Regardless of which scenario is being implemented, they all support the core services delivered by the platform: security, persistence, and connectivity etc. These services are provider-agnostic and run equally well in all use cases on multiple different Clouds providers.



The SAP Cloud Platform data centers are growing constantly. The figure, SAP Cloud Platform Data Centers - as of September 2019, shows a snapshot of the current data centers as of September 2019. See link for actual information <https://help.sap.com/doc/aa1ccd10da6c4337aa737df2ead1855b/Cloud/en-US/3b642f68227b4b1398d2ce1a5351389a.html>

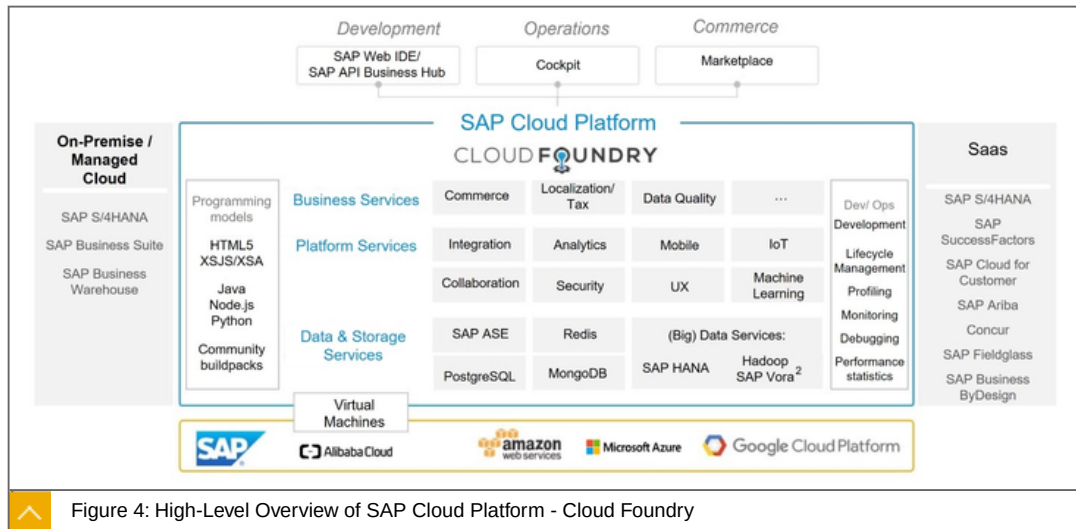


Figure 4: High-Level Overview of SAP Cloud Platform - Cloud Foundry

Initially, SAP Cloud Platform was delivered as Neo, a Java-based Cloud platform, which now co-exists with Cloud Foundry, which is a programming language agnostic alternative. This course mainly focuses on Cloud Foundry, rather than Neo, as this platform is the SAP preferred option for future Cloud usage. However, some services are not available on both Cloud platforms; Neo is currently utilized for the SAP Cloud Identity Provisioning Service (IPS) and the OAuth functionality, which we will discuss later in the course.

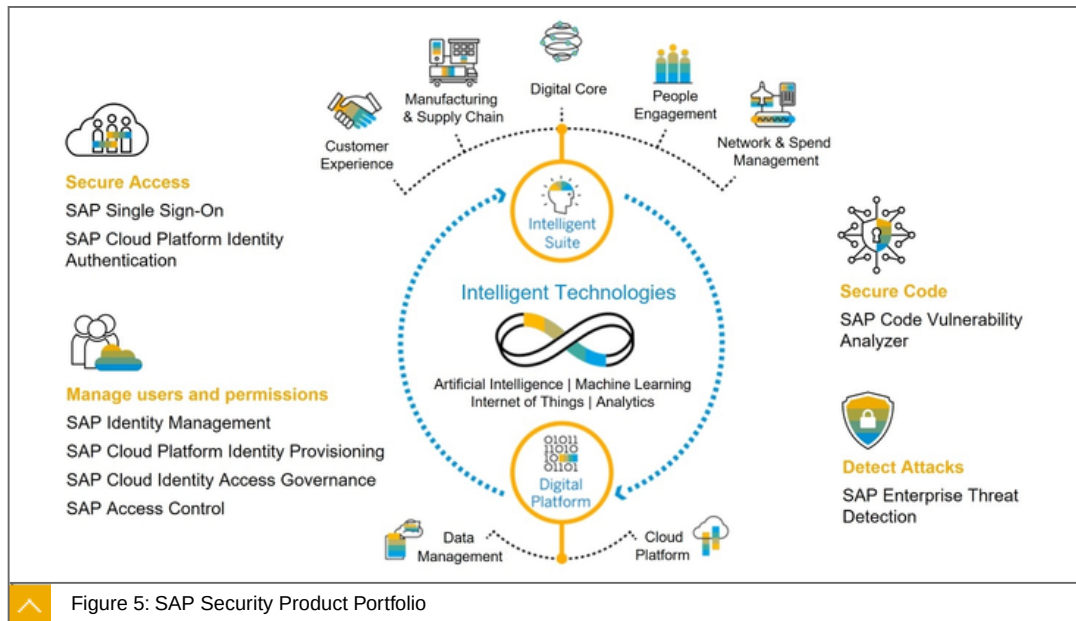


Figure 5: SAP Security Product Portfolio

The SAP security portfolio comprises from multiple Cloud and on-premise applications:

SAP Single Sign-On; provides simple, secure access to IT applications for business users. It offers advanced security capabilities to protect your company data and business applications.

SAP Cloud Platform Identity Authentication; provides simple and secure access to web-based applications with a variety of authentication methods anytime, from anywhere. The service was previously known as SAP Cloud Identity service.

SAP Identity Management; keeps user's data secure and consistent and supports customers by implementing integrated identity lifecycle scenarios with SAP's Cloud or on-premise HR solutions: SAP SuccessFactors (Cloud) and SAP ERP Human Capital Management (on-premise).

SAP IPS; offers a comprehensive, low cost approach to identity lifecycle management in the Cloud. IPS covers a broad range of source and target systems, both in the Cloud and on-premise.

SAP Cloud Identity Access Governance is a cloud solution that integrates out-of-the-box with SAP S/4 HANA and can run similar SOD (Segregation of Duties) scenarios as SAP GRC Access Control. Additionally it has functionalities to build business roles in the cloud, provision those to various target systems through SAP Cloud Platform Identity Provisioning and integrate in complex workflows thanks to SAP Cloud Platform Workflow service.

SAP GRC Access Control; helps to streamline the process of managing and validating user access to applications. SAP Identity Management and SAP Access Control can be used as an integrated solution for identity and access governance.

SAP Code Vulnerability Analyzer; carries out automated, static checks on your ABAP source code to identify security weaknesses.

SAP Enterprise Threat Detection; provides insight into malicious activities throughout IT landscapes and enables companies to identify security breaches in real-time. The solution detects internal and external attacks based on security event information in combination with context data.



LESSON SUMMARY

You should now be able to:

Describe SAP Cloud Platform and SAP Security Portfolio

Unit 1

Lesson 2

Describing the Security Setup in SAP Cloud Platform



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe the security setup in SAP Cloud Platform

Security Setup in SAP Cloud Platform

Cloud Cockpit Setup

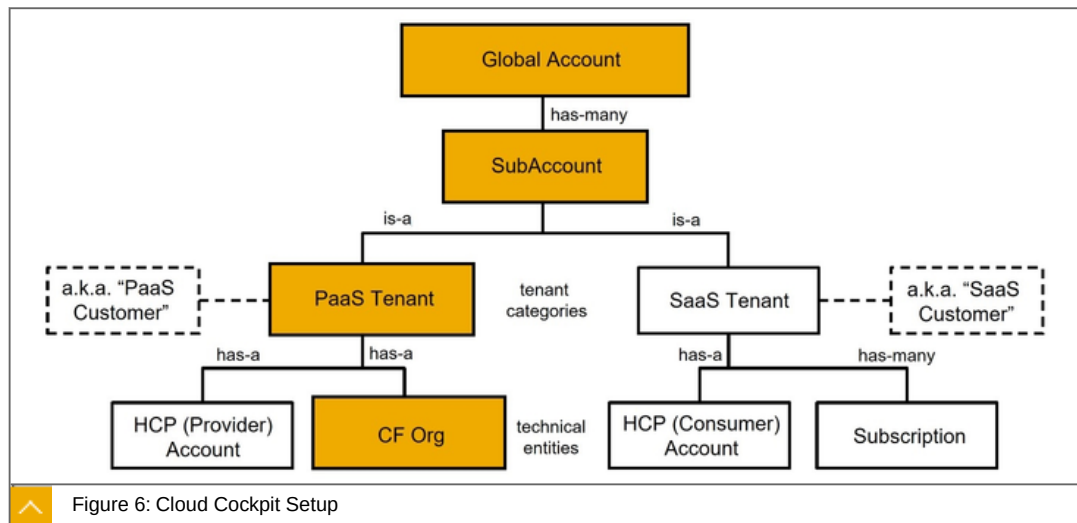
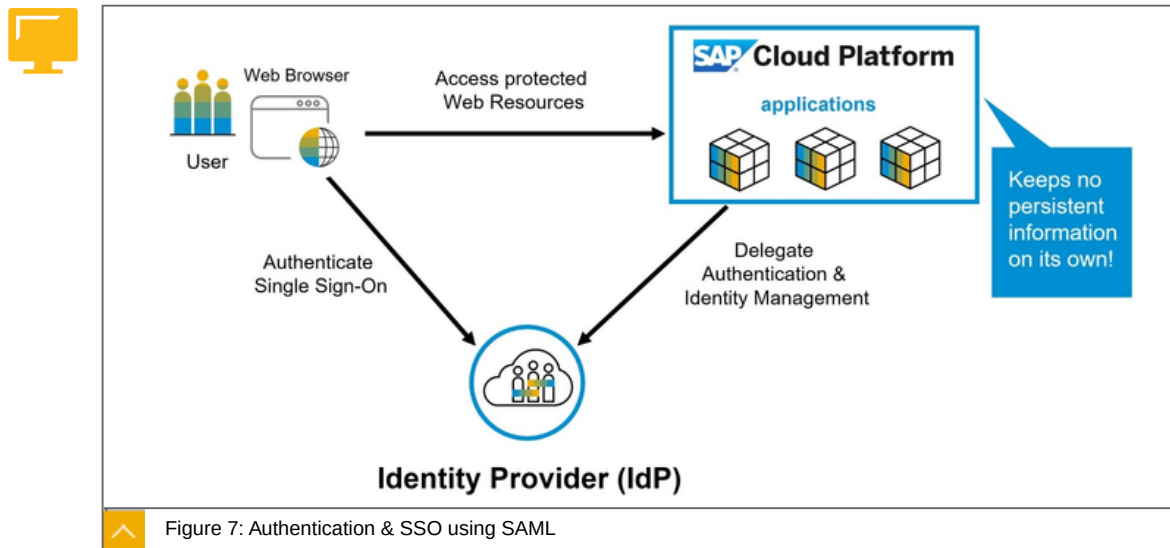


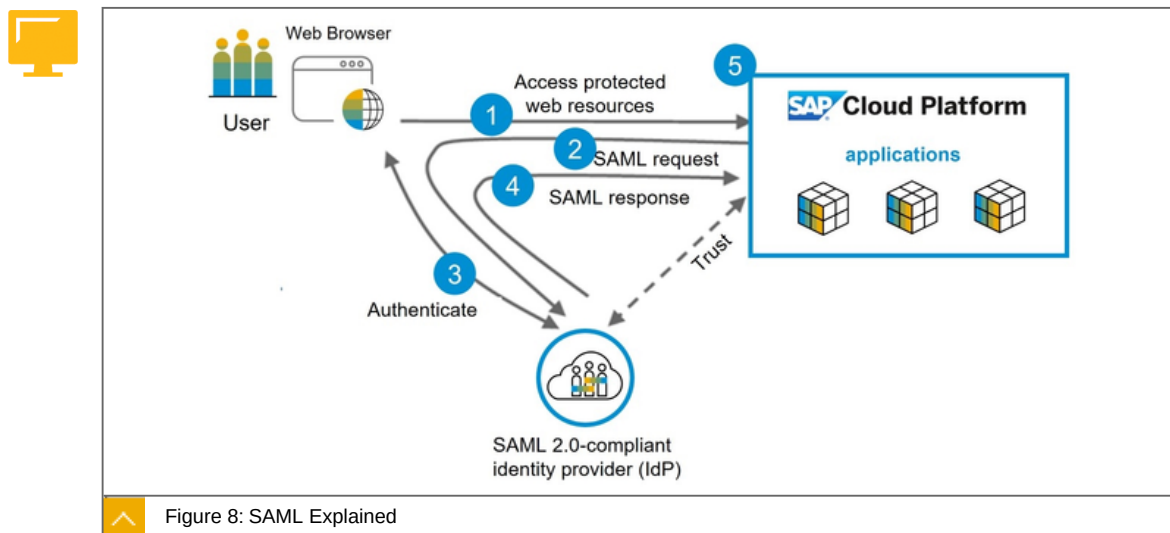
Figure 6: Cloud Cockpit Setup

The SAP Cloud Platform is organized in hierarchical accounts structure. A global account is assigned to a customer; there are multiple different types of subaccounts available within each global account. Apart from the infrastructure difference, a global account can have multiple Neo and/or Cloud Foundry subaccounts. On the next level down, each account can have multiple spaces, used for various developments and service consumptions.

Authentication & SSO using Security Assertion Markup Language (SAML)



SAML is the default authentication mechanism in the Cloud. The SAP Cloud Platform doesn't have an built-in Identity Provider (IdP) functionality, by default it uses the SAP standard IdP, called SAP ID. Customers can switch to a preferred IdP at any time or can have multiple IdPs for different scenarios.



For SAML flow to work, the two parties involved, the application hosting platform and the IdP, should have an established trust as per the following steps:

1. The user accesses the protected Web resource on SP.
2. The SAP Cloud application sends an SAML authentication request via HTTP, which is redirected to a trusted IdP.
3. The IdP authenticates the user, if it has not already been done.
4. Upon successful authentication, the IdP sends an HTML form with the SAML response message in a hidden field to the Web browser. This form contains the user's ID and

optional additional user attributes (for example, first name, last name, e-mail, group memberships, ...).

5. The user is authenticated based on information in the SAML response.

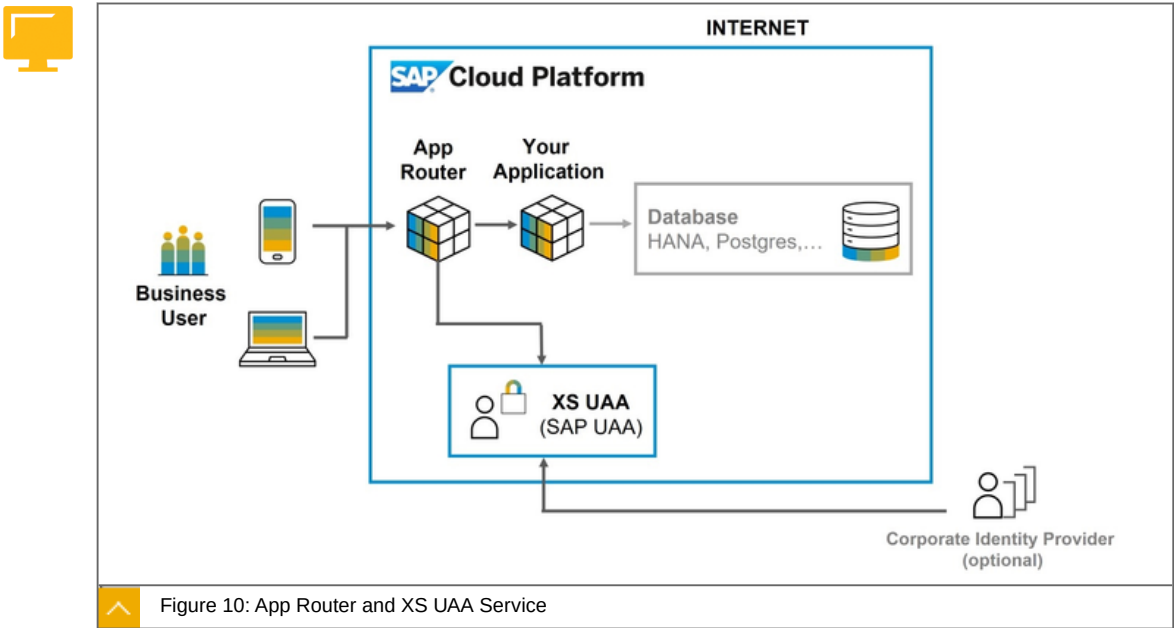
User Types in Relation to Supported IdPs



 Operators	Employees of platform/ Infrastructure provider	Corporate directory, VPN, 2 factor authentication
 Platform users	Customer: administrators, developers, support	SAP ID service SAP Cloud Platform Identity Authentication service (2 factor Authentication)
 Business users	End users of applications	Any SAML 2.0 compliant IdP

 Figure 9: User Types in Relation to Supported IdPs

App Router and XS UAA Service



The app router is an SAP-specific component. It orchestrates the app authentication and manages OAuth tokens. The SAP XS UAA service is instantiated via the subaccount and can reuse an already connected external IdP.

Roles and Scopes

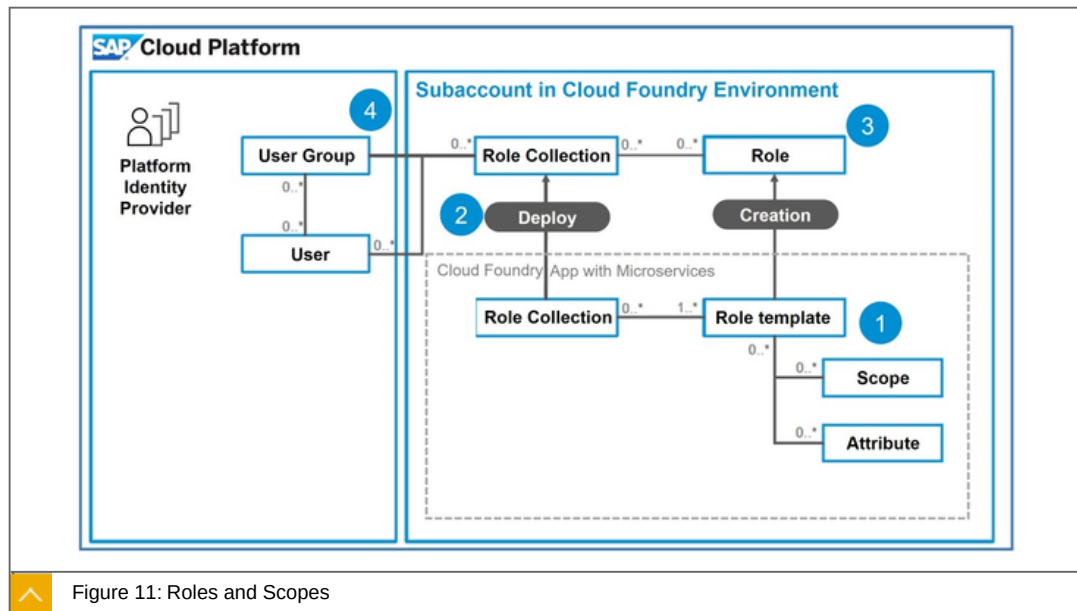


Figure 11: Roles and Scopes

An application deployed on the SAP Cloud Foundry can be protected using the following steps:

1. Maintain the xs-security.json with the correct scopes, attributes, role templates, and role collections.
2. Deploy the application to the SAP Cloud Platform. This will make the role collections available in the subaccount, where the application is deployed.
3. Create roles based on the role collection in case you need to maintain attribute values.
4. Maintain mapping between users and user groups in your IdP and maintain the mapping between the user groups from SAML and the role collections from your app, in your SAP Cloud Platform subaccount.

Users, Roles & Role Collections

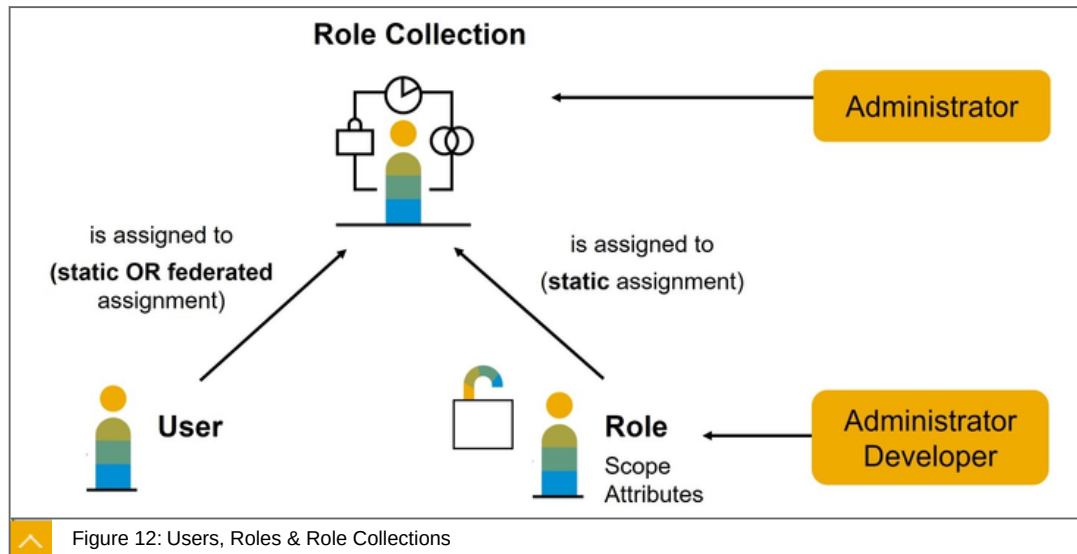


Figure 12: Users, Roles & Role Collections



```

{
  "xsappname": "sec161-teched-instance-based-auth",
  "scopes": [
    {
      "name": "${XSAPPNAME}.Edit",
      "description": "Edit scope"
    },
    {
      "name": "${XSAPPNAME}.Delete",
      "description": "Delete Scope"
    }
  ],
  "attributes": [
    {
      "name": "country",
      "description": "country",
      "valueType": "s"
    }
  ],
  "role-templates": [
    {
      "name": "Editor",
      "description": "Editor for Project Documentation",
      "scope-references": [ "${XSAPPNAME}.Edit", "${XSAPPNAME}.Delete" ],
      "attribute-references": [ "country" ]
    }
  ]
}

```



Developer

Role Template: "Editor"

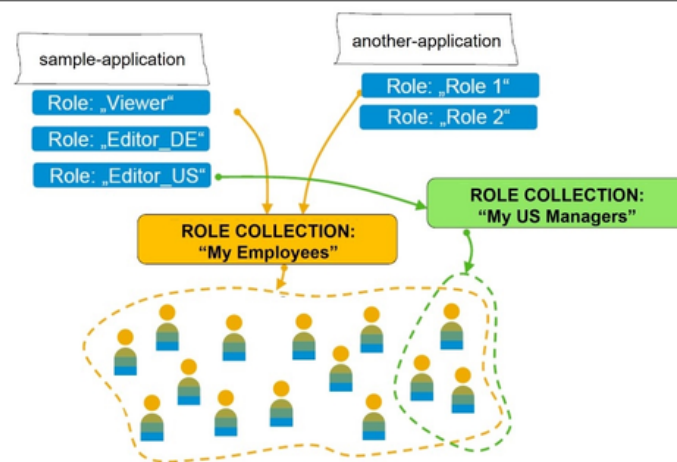
SCOPE: "Edit"

SCOPE: "Delete"

ATTRIBUTE: "Country"

Figure 13: Security in Design Time - xs-security.json

Design-time definition of scopes, attributes, and role templates.



An Administrator can define **role collections** using the administrative Uis

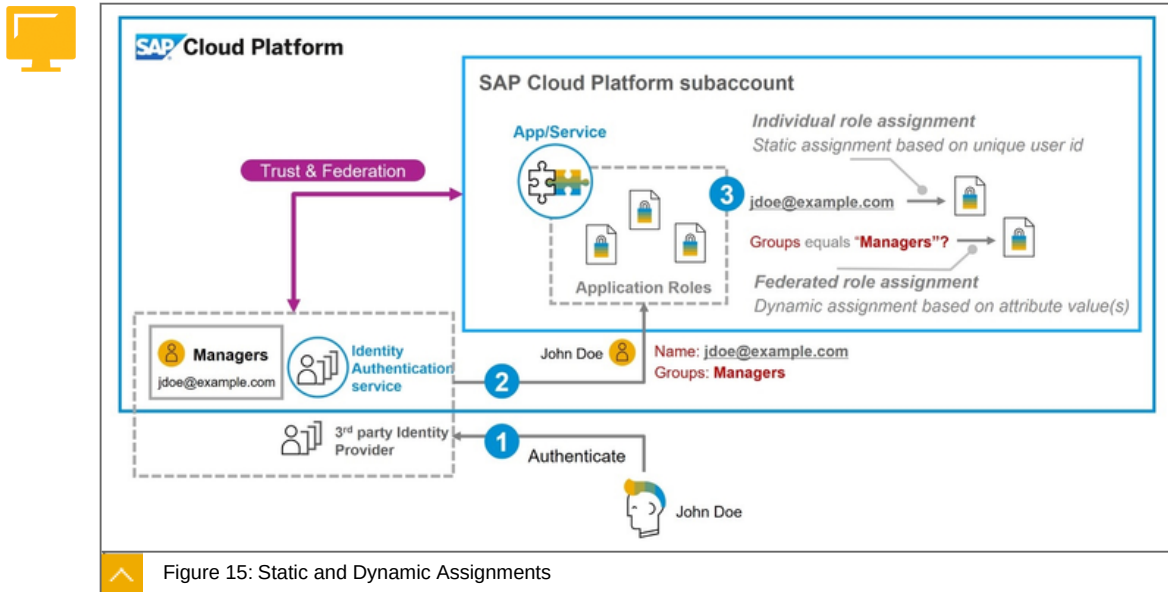
Multiple roles from different applications can be assigned to a single role collection

Using the CF Cockpit, a role collection can be **assigned** to a user or **mapped** to a **SAML Group**

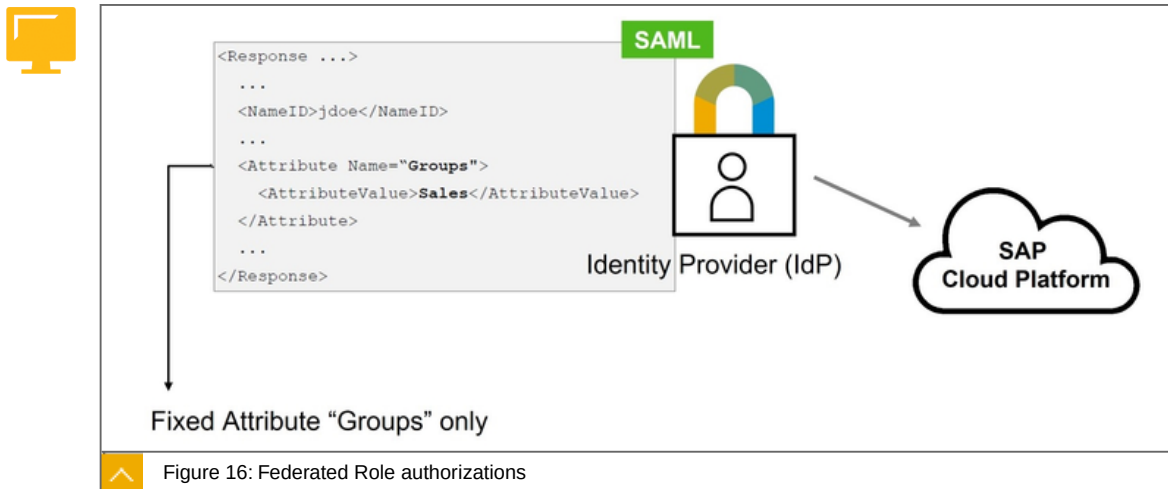
Figure 14: Assign Roles via Role Collections to Users

Role collections can contain roles from various applications.

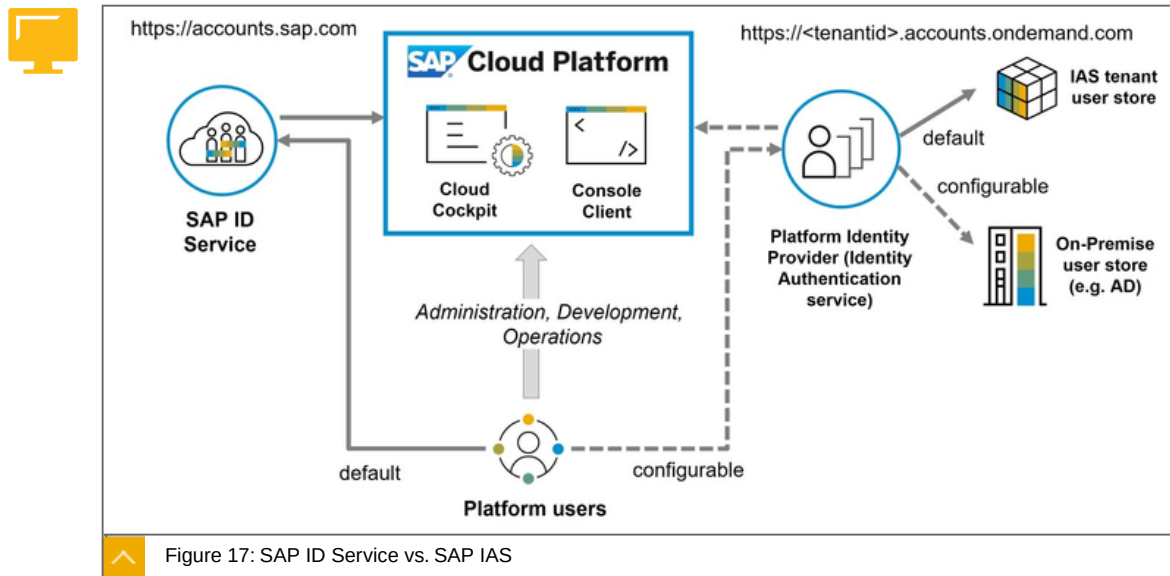
Static and Dynamic Assignments



Static assignments are done via the SAP Cloud Platform Cockpit. Dynamic assignments occur at runtime and are not saved in the system; they are valid until the user session expires. The default attribute for federated role assignment is called Groups (case sensitive).



SAP ID Service vs. SAP IAS



SAP ID Service	SAP IAS
S-User accounts	Full user and group lifecycle management
User base owned and managed by SAP	Integration with on-premise user store
Fixed password policy	Configurable password policy
Non-configurable MFA* for SAP CP Cockpit and console client access	Risk-based MFA*
	Customizable user interface
	...



LESSON SUMMARY

You should now be able to:

Describe the security setup in SAP Cloud Platform

Learning Assessment

1. What are the three main components of the digital enterprise according to SAP?

Choose the correct answer.

- ☐ **A** The Intelligent Suite, the Cloud Platform, and the Intelligent Technologies
- ☐ **B** The Intelligent Suite, the Digital Platform, and the Intelligent Technologies
- ☐ **C** Artificial Intelligence, Internet Of Things, and Machine Learning

2. Which security product from the SAP portfolio is purely Cloud based?

Choose the correct answers.

- ☐ **A** SAP Cloud Platform Authentication Service
- ☐ **B** SAP Cloud Platform Provisioning Service
- ☐ **C** SAP IdM

3. Which items are part of a role template?

Choose the correct answers.

- ☐ **A** Role collections
- ☐ **B** Scopes
- ☐ **C** Attributes

Learning Assessment - Answers

1. What are the three main components of the digital enterprise according to SAP?

Choose the correct answer.

- ☐ A The Intelligent Suite, the Cloud Platform, and the Intelligent Technologies
- ☒ B The Intelligent Suite, the Digital Platform, and the Intelligent Technologies
- ☐ C Artificial Intelligence, Internet Of Things, and Machine Learning

This is correct! The Intelligent Suite, the Digital Platform, and the Intelligent Technologies are the three main components of the digital enterprise in SAP.

2. Which security product from the SAP portfolio is purely Cloud based?

Choose the correct answers.

- ☒ A SAP Cloud Platform Authentication Service
- ☒ B SAP Cloud Platform Provisioning Service
- ☐ C SAP IdM

This is correct! SAP Cloud Platform Authentication Service and SAP Cloud Platform Provisioning Service are Cloud based.

3. Which items are part of a role template?

Choose the correct answers.

- ☐ A Role collections
- ☒ B Scopes
- ☒ C Attributes

This is correct! Scopes and attributes can be assigned to a role template.

UNIT 2

SAP Cloud Platform Identity Provisioning Service

Lesson 1

Understanding SAP Cloud Platform Identity Provisioning

16

UNIT OBJECTIVES

Understand SAP Cloud Platform Identity Provisioning

Unit 2

Lesson 1

Understanding SAP Cloud Platform Identity Provisioning



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Understand SAP Cloud Platform Identity Provisioning

SAP Cloud Platform Identity Provisioning Service (IPS)

IPS offers a comprehensive, low cost approach to identity lifecycle management in the Cloud.

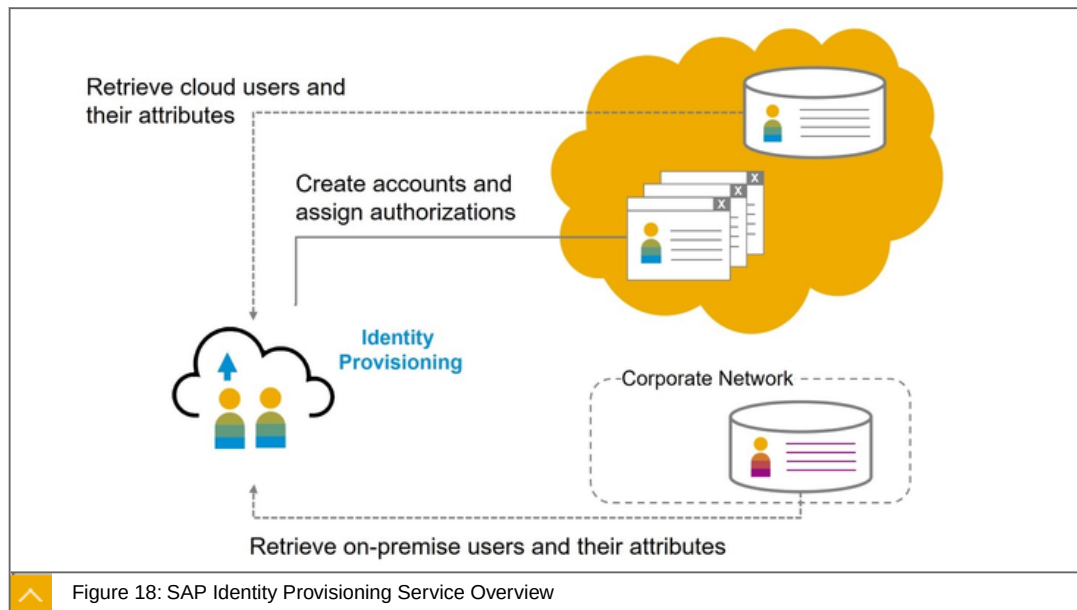


Figure 18: SAP Identity Provisioning Service Overview

Solution Overview

The IPS solution allows you to:

Manage user accounts and authorizations across Cloud and on-premise systems

Provision identities from user stores in the Cloud and on-premise

Enable business applications to quickly support single sign-on with identity authentication

Key Value Proposition

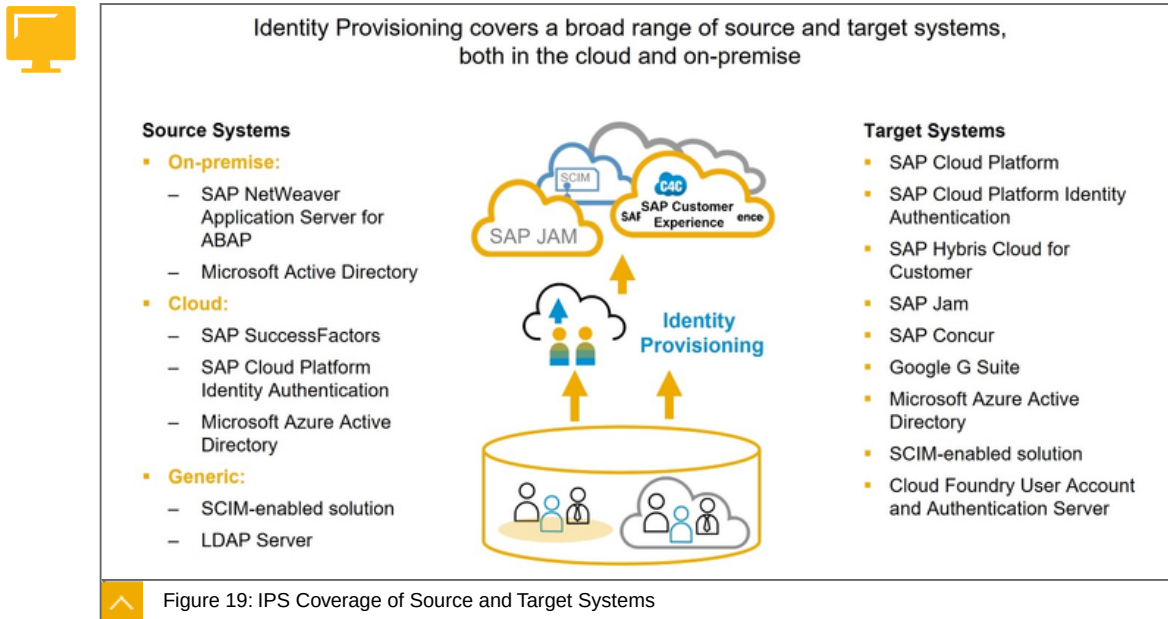
The IPS solution provides:

Fast and efficient administration of user on-boarding.

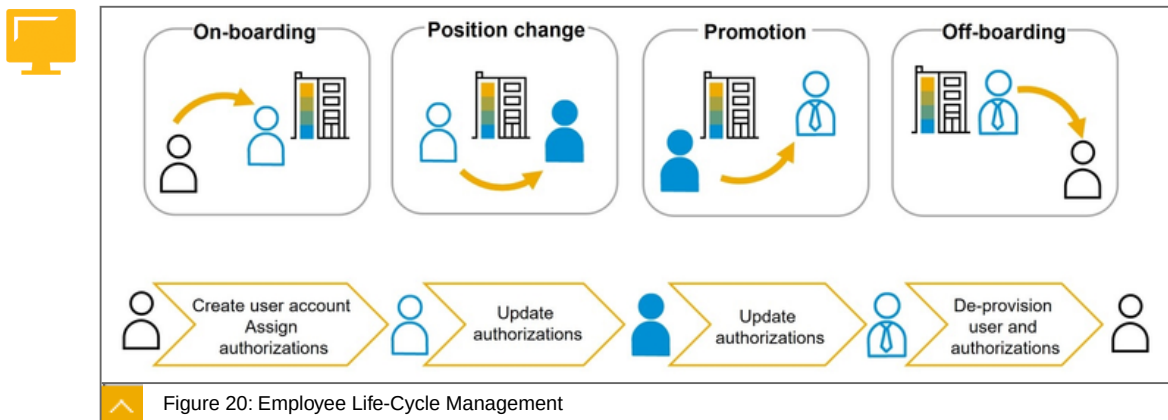
Centralized end-to-end lifecycle management of corporate identities in the Cloud.

Automated provisioning of existing on-premise identities to Cloud applications.

IPS Coverage of Source and Target Systems



Employee Life-Cycle Management



Efficient employee life-cycle coverage is critical for each enterprise, as it allows every point of the employee path to have the correct access and permissions.

SCIM - System for Cross-Domain Identity Management



- Current version – 2.0
- Standard protocol for accessing, querying, creating updating and deleting identity information
- Modeled REST endpoints for the various resource types (defined in RFC7644)
- Each resource must have an id, externalId and meta as attributes

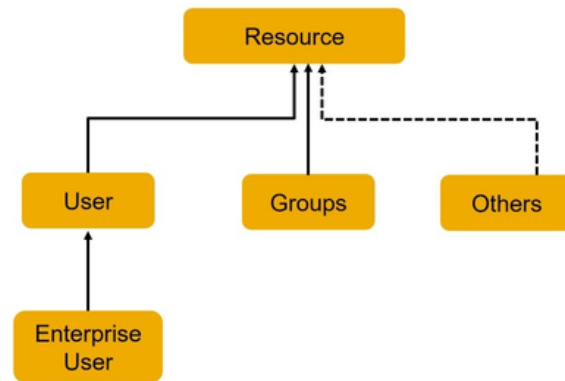


Figure 21: SCIM - System for Cross-Domain Identity Management

The System for Cross-domain Identity Management (SCIM) specification is designed to manage user identity in Cloud-based applications and services in a standardized way which enables interoperability, security, and scalability. The goal of SCIM is to automate the identity provisioning process, therefore, reducing the cost of identity management and improving security. The automation process simplifies the user experience.

The underlying concept of SCIM 2.0 is based on a common user schema, group schema, and an extension model which are exchanged via an HTTP-based protocol. It has an object model where a Resource is the common denominator and all the other SCIM objects are derived from it. A resource must have an id, externalId, and meta as attributes. User, Group and EnterpriseUser extend the common attributes. SCIM is not exclusive to these resources and may be extended to support other resource types.

A SCIM server provides a set of resources, the allowable contents of which are defined by a set of schema URIs and a resource type. A Resource is a JSON object containing a schema attribute that indicates a list of one or more URIs included in the SCIM schemas. An attribute consists of the attribute name and one simple or complex value, either of which may be multi-valued.

Example of a User Resource

```

{
  "schemas": ["urn:ietf:params:scim:schemas:core:2.0:User"],
  "id":
    "2819c223-7f76-453a-919d-413861904646",
  "externalId": "701984",
  "meta": {
    "resourceType": "User",
    "created":
      "2010-01-23T04:56:22Z",
    "lastModified":
      "2011-05-13T04:42:34Z",
    "version": "WWa330bc54f0671c9\\",
  }
}
  
```



```

"location":
  "https://example.com/v2/Users/2819c223..."
},
"userName":
  "ccampbell@example.com",
"name":
  "familyName": "Campbell",
  "givenName": "Chris",
  "honorificPrefix": "Mr."
},
"displayName": "Chris Campbell",
"profileUrl": "https://login.example.com/ccampbell",
"emails":
  [{
    "value": "ccampbell@example.com",
    "type": "work",
    "primary": true
  }, {
    "value": "ccampbell@sap.com",
    "type": "home"
  }
],
"groups": [{
  "value":
    "e9e30dba-f08f-4109-8486-d5c6a331660a",
    "$ref": "https://example.com/v2/
    Groups/e9e30dba...",
    "display": "SECCL1 Auditors"
  }
]
}

```

The supported set of operations are as follows:

GET -> Retrieve one or more complete or partial resources

POST -> Depending on the endpoint can create new resources, for example, User, Search, Bulk, etc.

PUT -> Modifies a resource by replacing existing attributes with a specified set of replacement attributes

PATCH -> Modifies a resource with a set of specific changes (partial update)

DELETE -> Deletes a resource

As a REST-based protocol, every SCIM resource has a URL. This URL has three components:

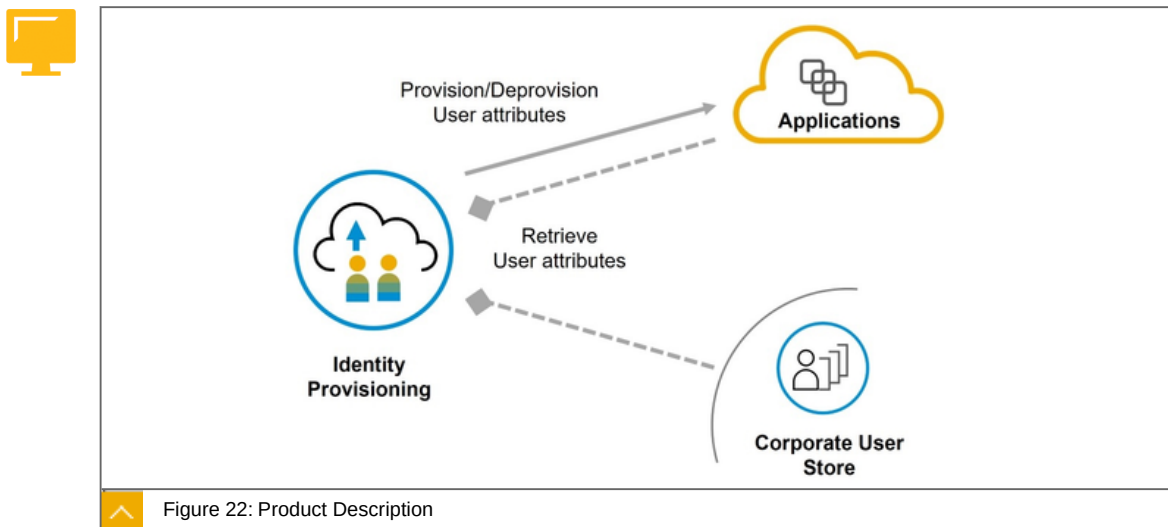
The SCIM server based URL, which includes the protocol scheme (in production environments, this should be: https://, the hostname, and port if necessary), and the base path of the SCIM endpoint (this will vary from product to product; for the OIM SCIM server, it is /idm/scim/v1)

The resource type, for example, Users

An identifier of the specific resource, which is usually a number or a UUID; SCIM clients should not make any assumptions about the structure of SCIM resource identifiers

An example is: <https://myserver.example.com/idm/scim/v1/Users/136971>

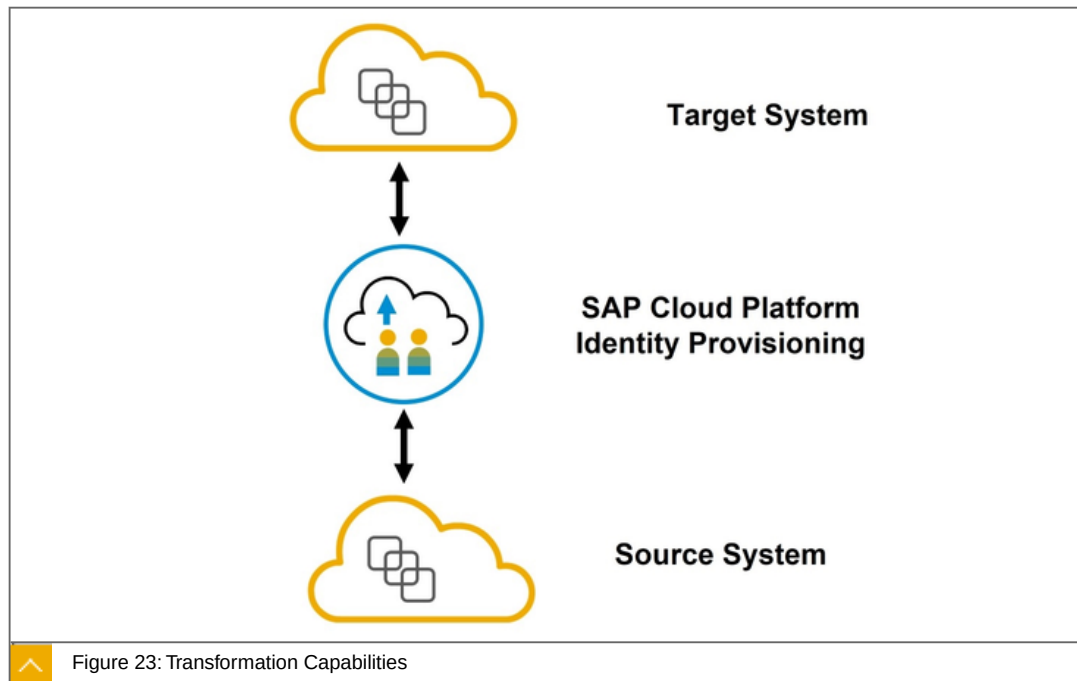
Product Description



Identity and Access Management for the Cloud and hybrid is a service of the SAP Cloud Platform. It provides identity lifecycle management for Cloud-based business applications. It is integrated with SAP Identity Management (IdM) for hybrid landscapes and also non-SAP IdM solutions using the SCIM standard.

SAP IdM is a simple and agile solution with a short time-to-value. It provides simple and agile on-boarding of users and applications.

Transformation Capabilities



Transformation Capabilities

Policy-based assignments

- Define rules for assignments based on the input data
- For example, they use the value of an identity's unit to decide on the required roles

Mapping between identity models

- Map between attributes in different models, for example, surname to family name
- Adjust the data format, for example, for time or number formats

Filtering

- Supports the sorting of objects that should be read or written in detail



LESSON SUMMARY

You should now be able to:

Understand SAP Cloud Platform Identity Provisioning

Learning Assessment

1. What can be provisioned with IPS using SCIM?

Choose the correct answer.

- ☐ **A** Groups
- ☐ **B** Assignments
- ☐ **C** User Attributes
- ☐ **D** Users

Learning Assessment - Answers

1. What can be provisioned with IPS using SCIM?

Choose the correct answer.

- ☒ **A** Groups
- ☒ **B** Assignments
- ☒ **C** User Attributes
- ☒ **D** Users

This is correct! Groups, Assignments, User Attributes, and Users can be provisioned by IPS using SCIM.

UNIT 3

SAP Identity Management

Lesson 1

Describing SAP Identity Management

25

UNIT OBJECTIVES

Describe SAP Identity Management

Unit 3

Lesson 1

Describing SAP Identity Management



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe SAP Identity Management

SAP Identity Management (IdM)

SAP IdM maintains user data securely and consistently; it supports customers by implementing integrated identity lifecycle scenarios with SAP's Cloud or on-premise HR solutions: SAP SuccessFactors solutions (Cloud) and SAP ERP Human Capital Management (on-premise).



- Integrate business processes
- Improve productivity
- Improve security



Figure 24: One Central source of Truth for Identity Data

SAP IdM provides the following functionality:

Integrated business processes

- Integration with SAP SuccessFactors solutions and SAP ERP Human Capital Management automates user access provisioning based on current business roles
- Built-in compliancy in the process integrates with SAP Access Control for segregation of duty conflict analysis

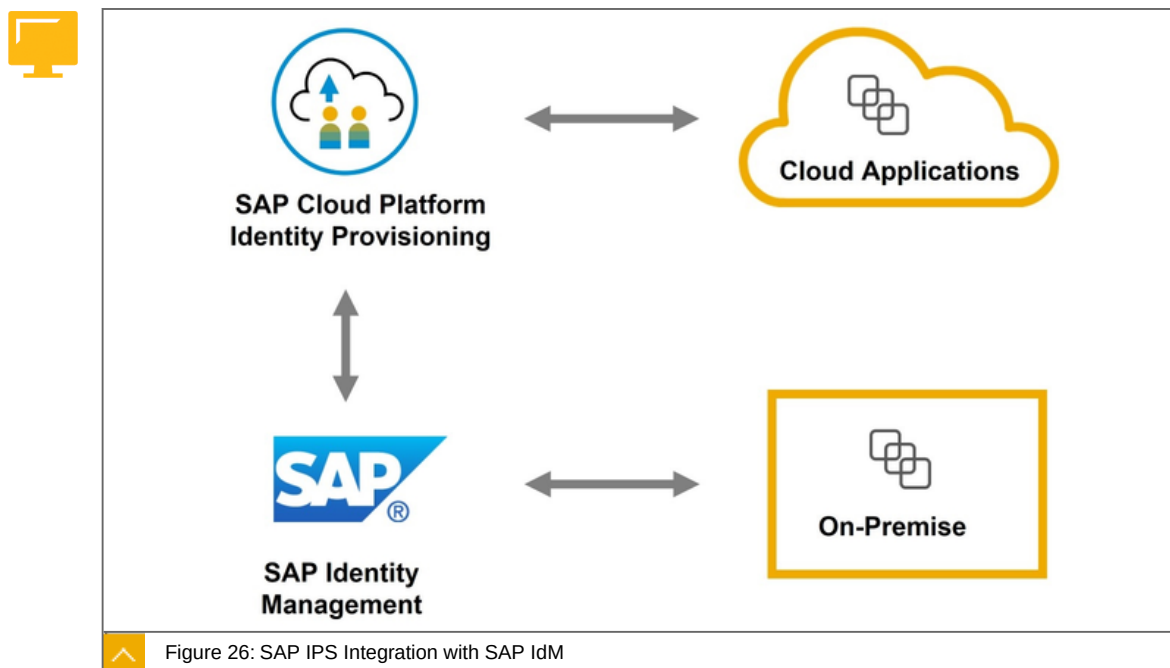
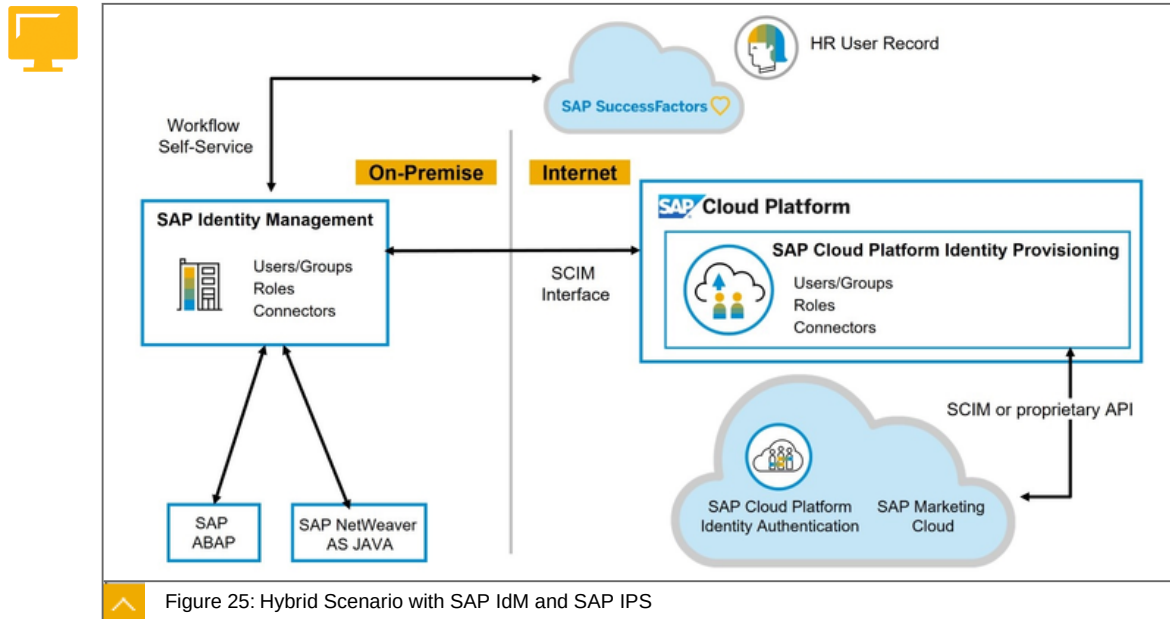
Improved productivity

- Self-service for end-user password reset
- Self-service for end-user access request

Improved security

- Reduces risk with centralized user IdM across SAP and non-SAP solutions
- Automatically revokes or adapts access when a person leaves a company or in case of an internal move

Hybrid Scenario with SAP IdM and SAP IPS



SAP IdM is recommended for on-premise landscapes as it is optimized for on-premise applications in terms of customization and performance etc.

Identity provisioning is recommended for Cloud systems as it:

Offers a deployment model and simplicity suitable for Cloud-based business applications.

Allows customers to efficiently on-board new applications.

Hybrid scenarios recommendations:

Customers of SAP IdM can extend their identity lifecycle management to the Cloud using identity provisioning.

Integration of SAP IdM with identity provisioning allows customers to benefit from the advantages of both methods.

The benefits provided are as follows:

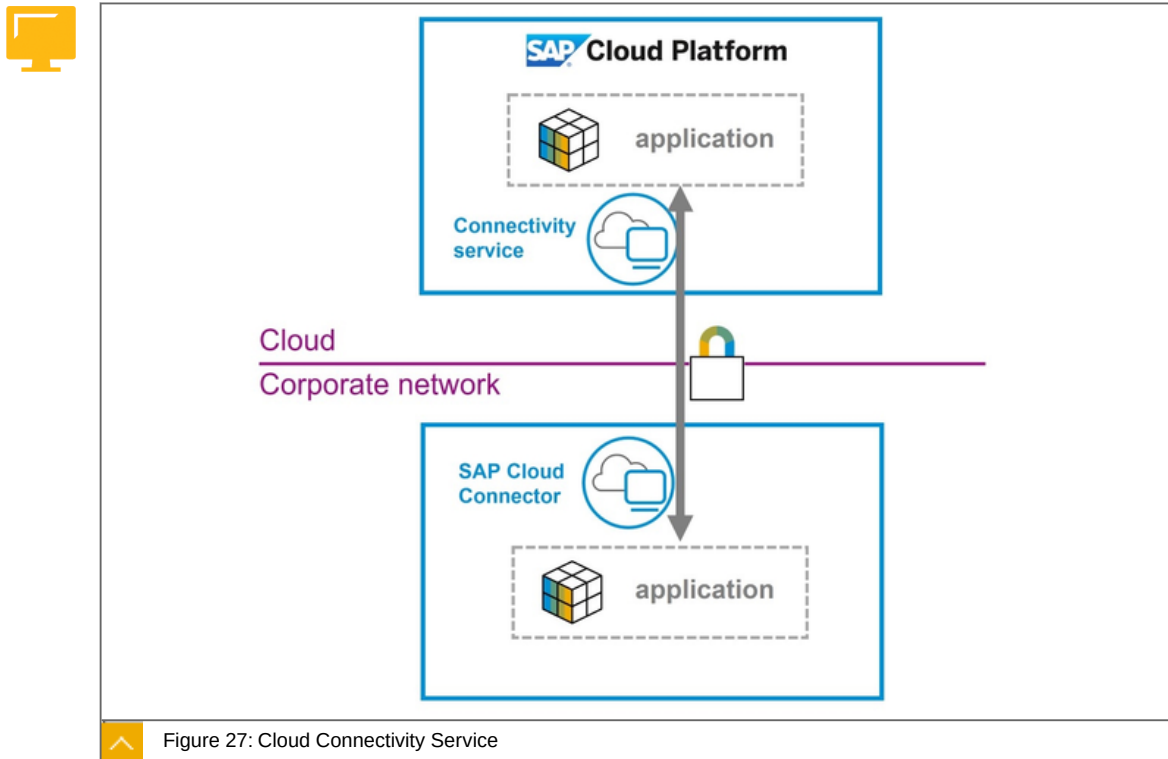
An existing SAP IdM installation is not affected when enabling hybrid identity management.

Cloud-based business applications become available as supported systems in SAP IdM.

SAP IdM capabilities, such as, self-service, workflows, or business role management are still available even for the hybrid scenarios.

SAP IdM capabilities for reporting and auditing purposes are reused to cover the Cloud.

Cloud Connectivity Service



Cloud Connectivity Services

Cloud connectivity provides the following services:

It establishes secure VPN connection between the SAP Cloud Platform and on-premise systems

There are no required changes in the existing corporate firewall configuration

It initiates encrypted connections to Cloud application from inside the on-premise network to the Cloud

It secures access to on-premise systems, include the following:

- Fine grained access control lists of allowed on-premise resources
- Fine grained audit logging for traceability
- Principal propagation from Cloud to on-premise
- Trust relation with on-premise system based on X.509 certificates

The Cloud connector supports the following communication protocols:

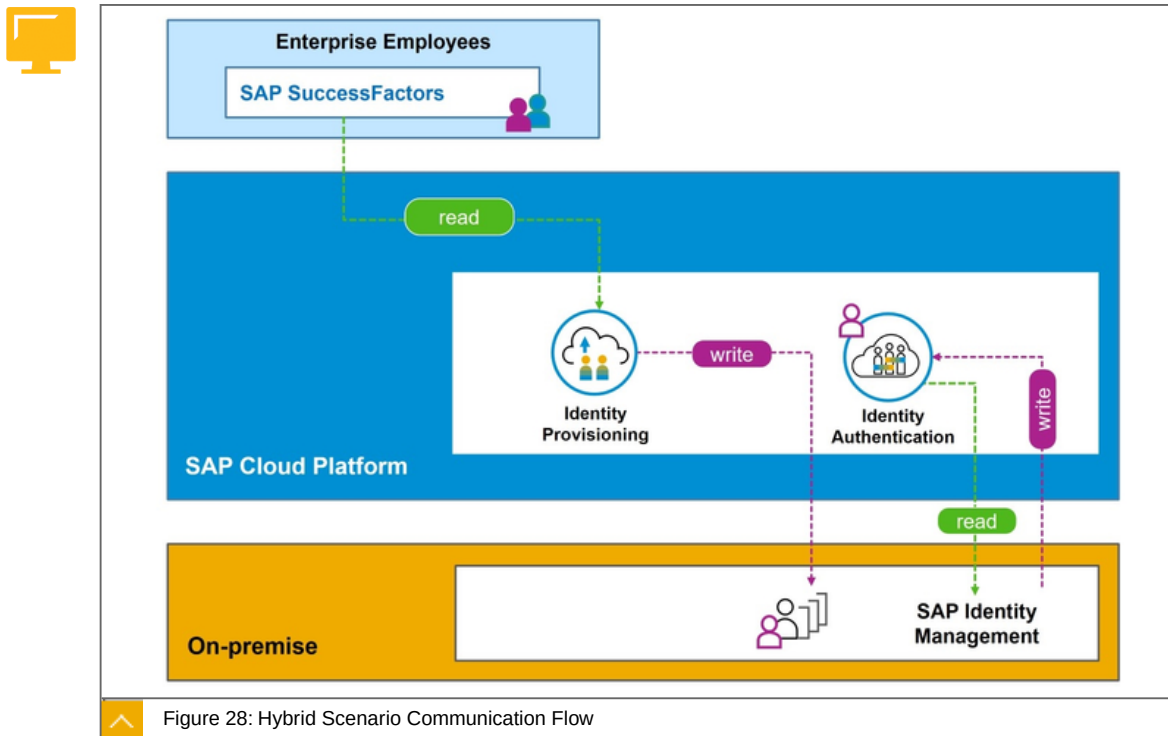
HTTP(s)

RFC

LDAP

TCP

Hybrid Scenario Communication Flow



LESSON SUMMARY

You should now be able to:

Describe SAP Identity Management

Learning Assessment

1. Which of the following SAP products are involved in the hybrid scenario?

Choose the correct answer.

- ☐ **A** SuccessFactors, Identity Authentication, and IdM
- ☐ **B** S/4HANA on-premise, IPS, and IAS
- ☐ **C** SuccessFactors, IPS, and IdM

2. Which of the following communication protocols are supported by SAP Cloud Connector?

Choose the correct answers.

- ☐ **A** LDAP
- ☐ **B** HTTP
- ☐ **C** HTTPS

Learning Assessment - Answers

1. Which of the following SAP products are involved in the hybrid scenario?

Choose the correct answer.

- ☐ A SuccessFactors, Identity Authentication, and IdM
- ☐ B S/4HANA on-premise, IPS, and IAS
- ☒ C SuccessFactors, IPS, and IdM

This is correct! The SAP products involved in the hybrid scenario are SuccessFactors, IPS, IdM.

2. Which of the following communication protocols are supported by SAP Cloud Connector?

Choose the correct answers.

- ☒ A LDAP
- ☒ B HTTP
- ☒ C HTTPS

The protocols supported by SAP Cloud Connector are LDAP, HTTP, and HTTPS.

UNIT 4

SAP Cloud Platform Identity Authentication Service

Lesson 1

Understanding SAP Cloud Platform Identity Authentication Service

32

UNIT OBJECTIVES

- Understand SAP Cloud Platform Identity Authentication and Usage Scenarios
- Configure Branding and Customization
- Configure Authentication Options

Unit 4

Lesson 1

Understanding SAP Cloud Platform Identity Authentication Service



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand SAP Cloud Platform Identity Authentication and Usage Scenarios
- Configure Branding and Customization
- Configure Authentication Options

SAP Cloud Platform Identity Authentication Service (IAS)

SAP Cloud Platform IAS provides simple and secure access to Web based applications with a variety of authentication methods at anytime and from anywhere. The service was previously known as SAP Cloud ID service.

Authentication and Single Sign-On in the Cloud

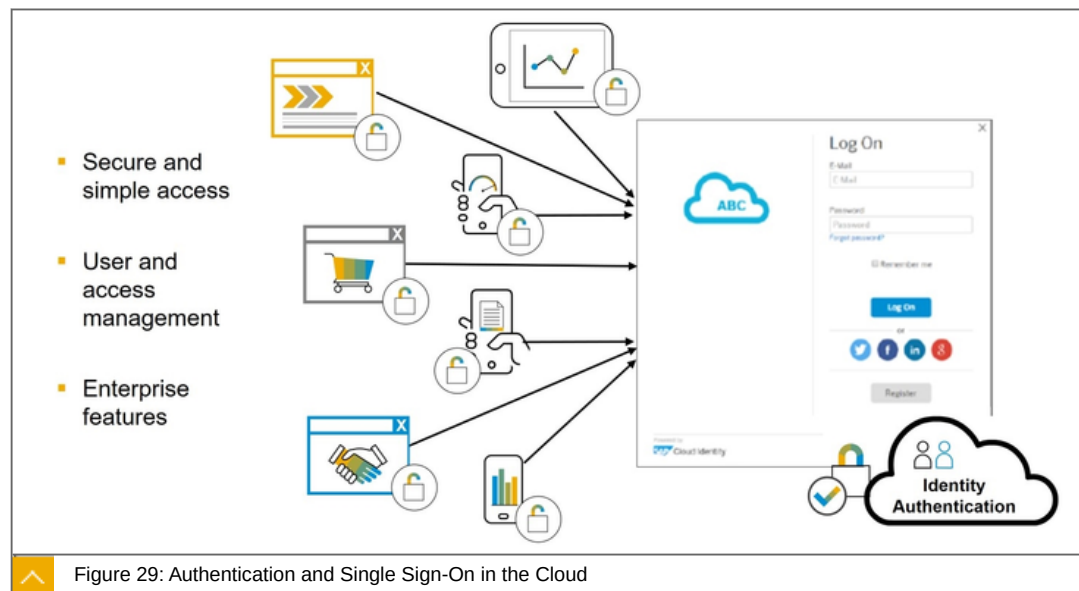


Figure 29: Authentication and Single Sign-On in the Cloud

IAS provides secure and simple access based on the following factors:

- Identity federation based on SAML 2.0
- Web Single Sign-On SSO and desktop SSO
- Secure on-premise integration to reuse existing authentication systems

Social login and two-factor authentication

Risk-based authentication

IAS provides user and access management based on the following factors:

User administration and integration with on-premise user stores

User groups and application access management

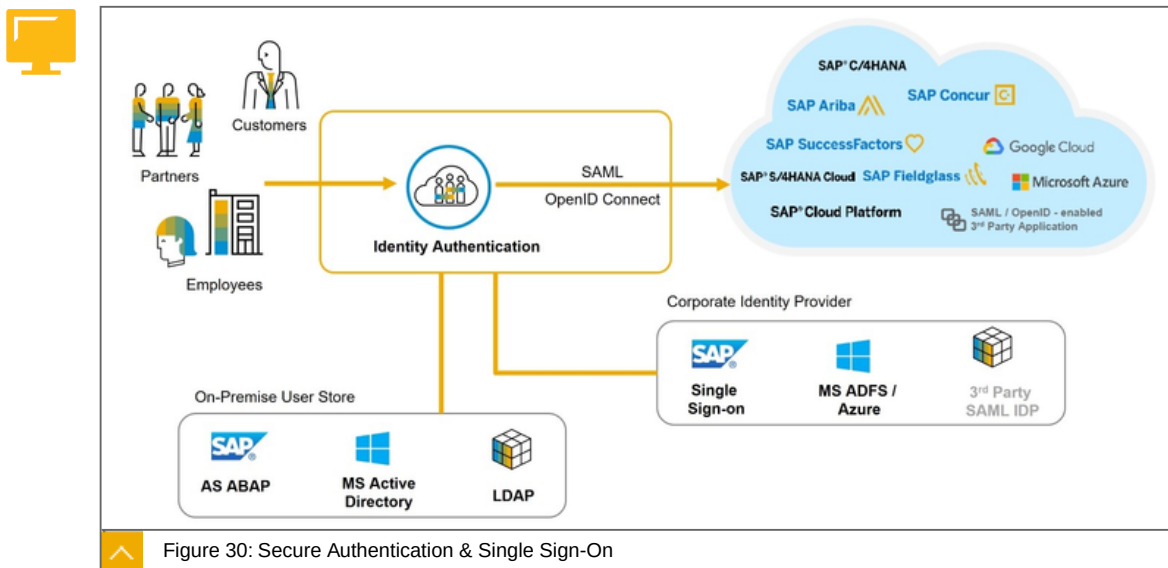
User self-service, for example, password reset, registration, and user profile maintenance

System for Cross-domain Identity Management (SCIM) API

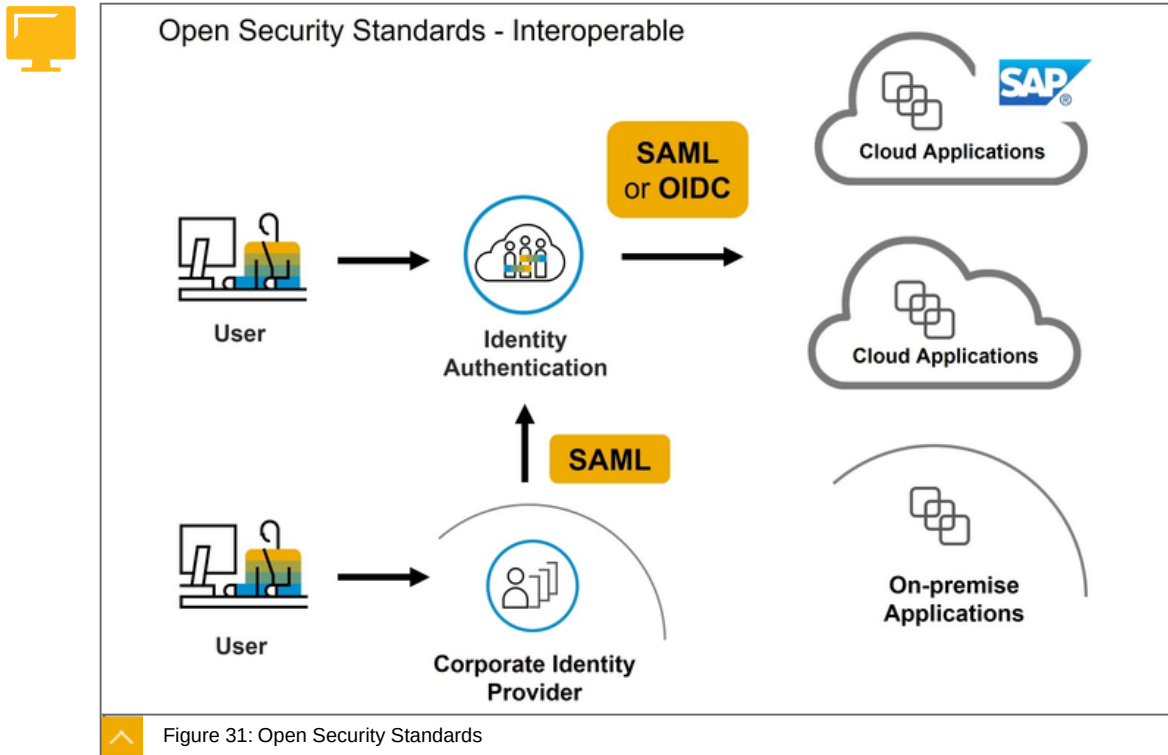
IAS provides the following enterprise features:

Branding of end user UIs

Password and privacy policies

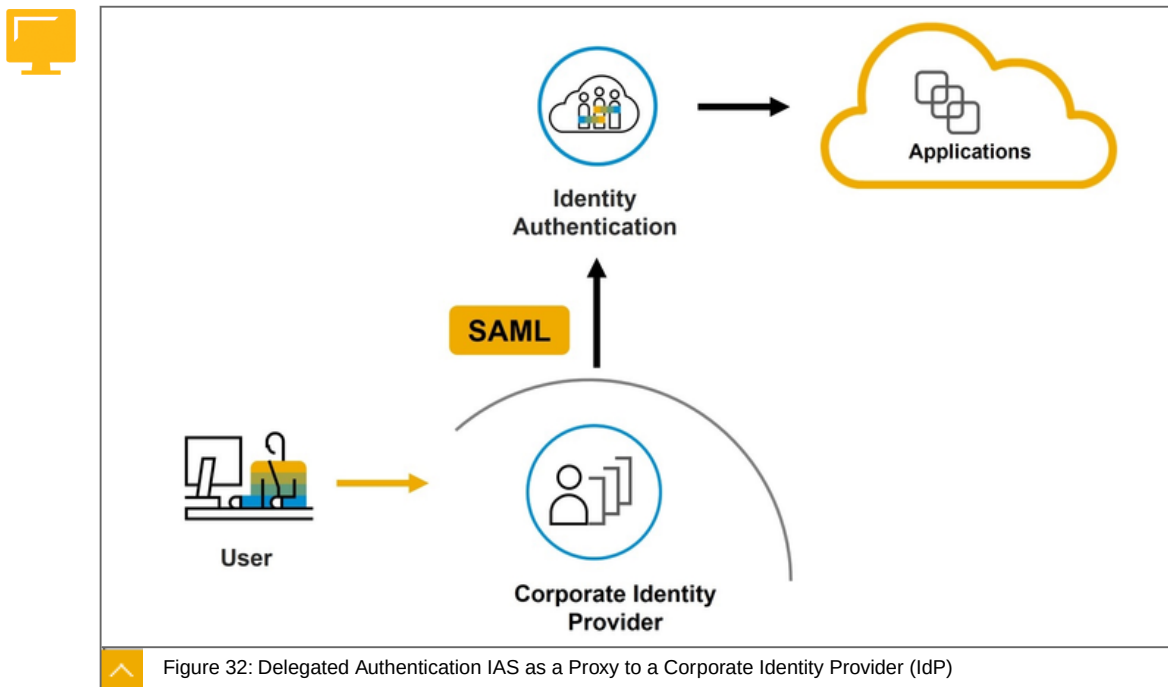


Open Security Standards



IAS is interoperable with all application supporting SAML* 2.0 standard or OpenID Connect (OIDC).

Delegated Authentication IAS as a Proxy to a Corporate Identity Provider (IdP)



IAS has the following IdP proxy features:

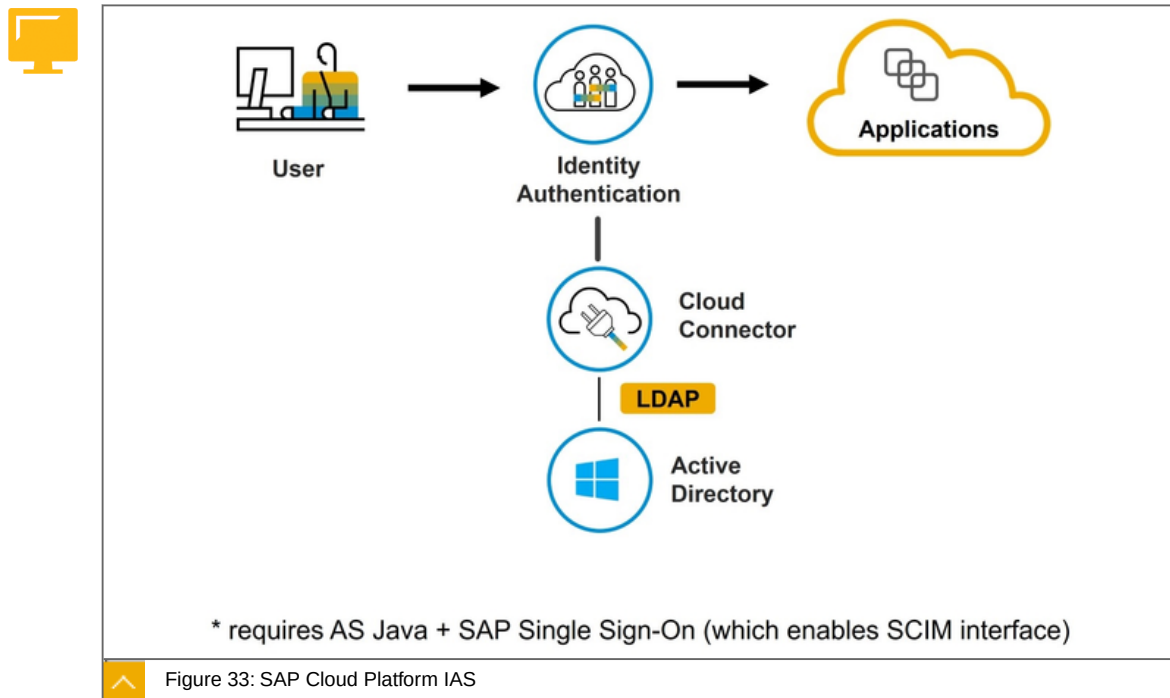
Authentication is delegated to corporate IdP login

Reuse of existing SSO infrastructure

Easy and secure authentication for employee scenarios

Federation based on the SAML 2.0 standard

Delegated Authentication - Authentication with an On-Premise User Store



IAS can connect to an on-premise user store.

Users credentials are taken from:

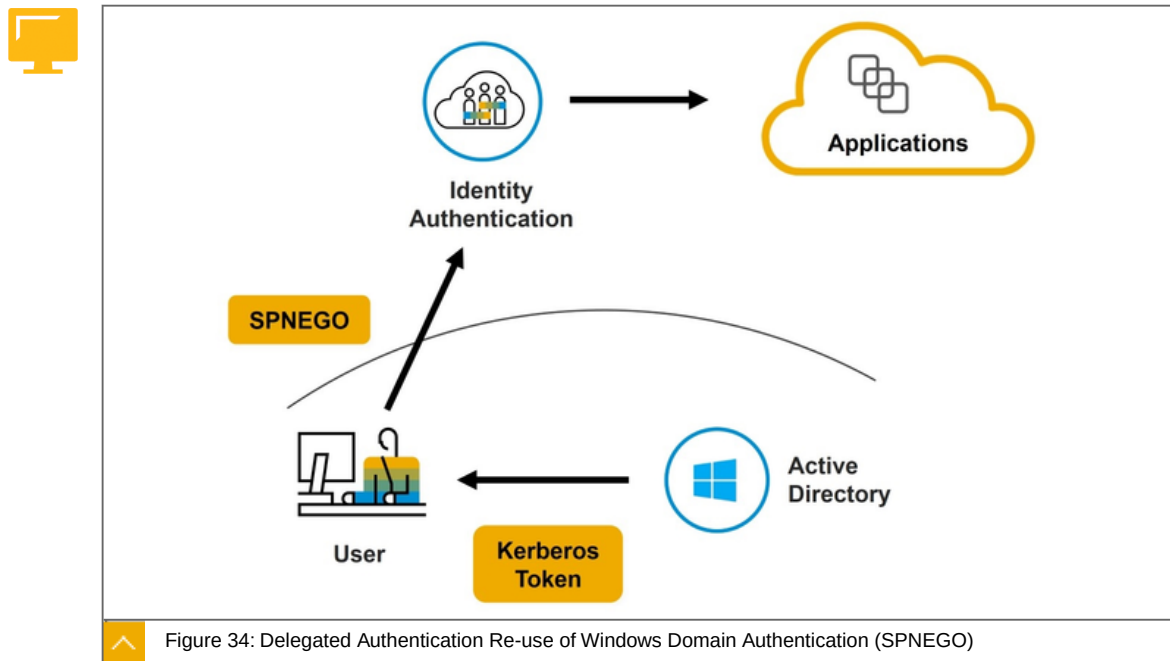
- Active Directory (through LDAP)
- AS Java (which can be either local UME, ABAP store or AD)

There is no user replication required to the Cloud

Internal network ports do not need to be exposed to the Internet

Other IAS product features can be used including UI configuration policies and two-factor authentication

Delegated Authentication Re-use of Windows Domain Authentication (SPNEGO)



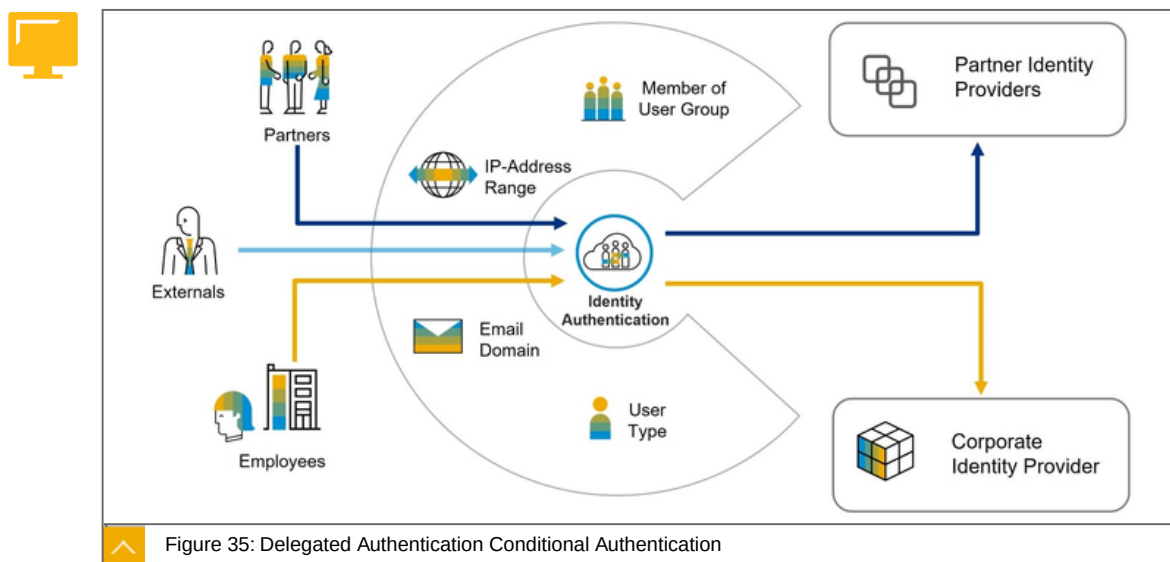
SPNEGO authentication provides the following:

Users authenticated with Microsoft Active Directory can utilize SSO for Cloud applications without re-authentication

Reuse of existing corporate identity infrastructure

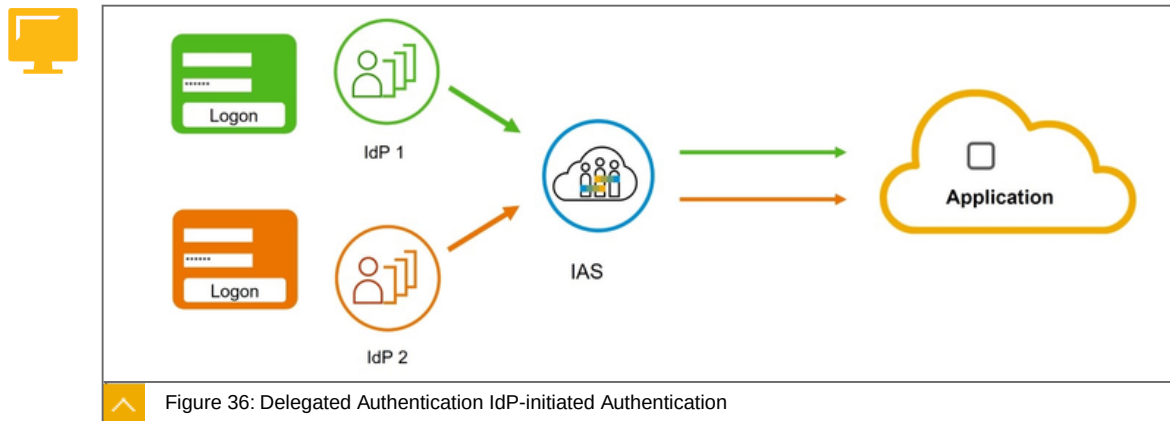
Secure authentication and SSO for Cloud and on-premise Web applications

Delegated Authentication Conditional Authentication



Depending on several factors, different types of users can be re-routed to different IDPs for authentication.

Delegated Authentication IdP-initiated Authentication



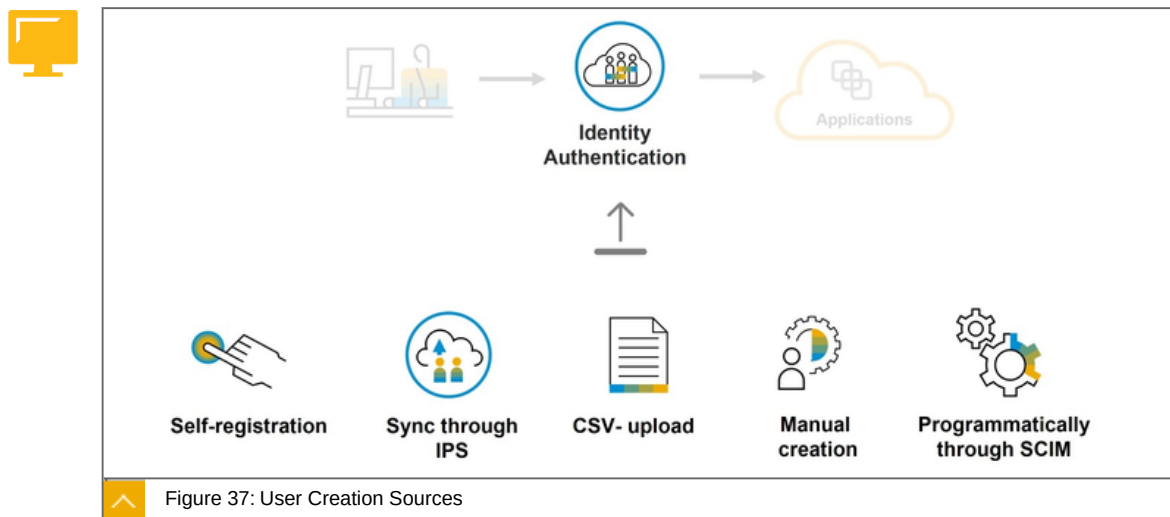
As a proxy to multiple IdPs, IAS provides:

A secure business network and allows partner users to login via their corporate IdP

Authentication that is initiated by the corporate IdP

An optional check for correct user group assignment can be configured upon successful authentication; a sync of users from IdPs to groups in IAS is required

User Creation Sources



Branding and Customization



Customization Features

- Company Logo
- Application Name and Logo
- Color style
- Terms of Use & Privacy Policy
- Adjust UI Texts via API
- eMail Templates

Product Features

- Responsive UIs
- Multilanguage support

Log On

E-mail
E-Mail

Password
Password

☐ Remember me

Log On

Register

Forgot password?

Terms and Conditions

☐ I have read the [Privacy Statement](#) and consent to this agreement. *

☐ I have read and understood the Terms and Conditions of Document Center. *

*Required

Register

Figure 38: Branding and Customization

CONCUR

Welcome

If you experience difficulty logging into the site, please click on the [forgot your password?](#) link.

Log In

User Name

Password

☐ Remember user name on this computer

Forgot your user name? [Forgot your password?](#)

SAP Jam

Log in to SAP Jam

Email Address

Password

☐ Remember me

Forgot Password? [Yes](#)

SAP Ariba Proposals

Powered by India Sourcing

Supplier Login

User Name

Password

Log In

Forgot Username or Password

Log On

E-Mail

E-Mail

Password

Password

☐ Remember me

Log On

User

Password

Language EN - English

Client 100

Log On

Change Password

Figure 39: Beautiful Logon Screens

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User Self-Services

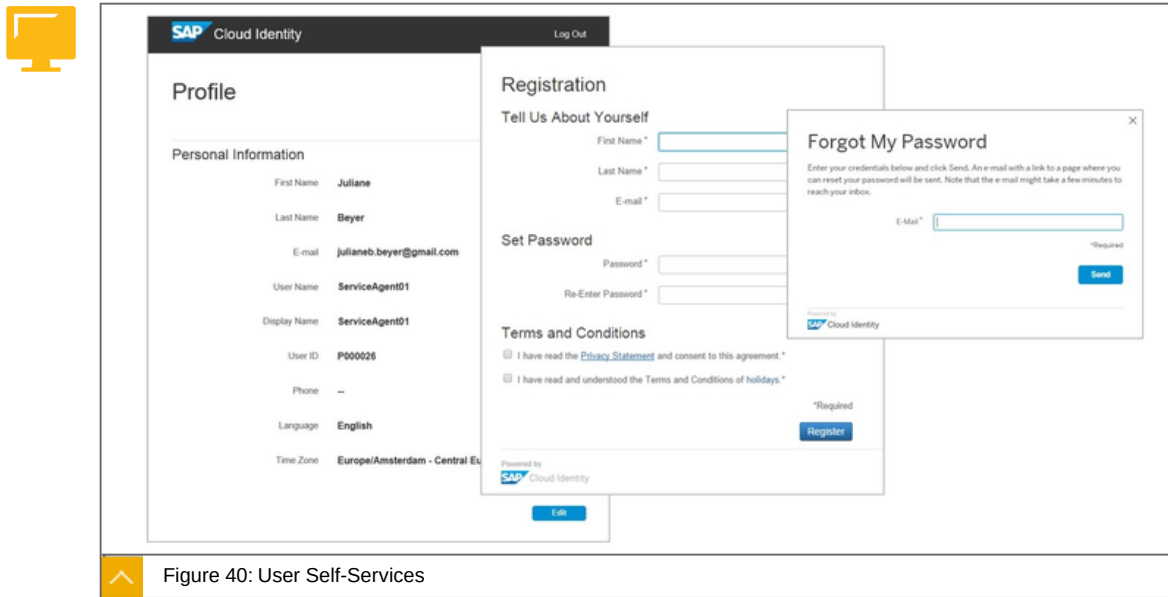


Figure 40: User Self-Services

IAS provides convenient user self-service that includes the following features:

- Self-registration
- Account confirmation via email
- Forgotten password reset
- Edit of account details or password change
- Activation of 2-factor authentication
- Linking or unlinking of social accounts

Access Protection for Applications



Access Protection for Applications

Registration

Tell Us About Yourself

First Name *

Last Name *

E-Mail *

Phone *

Set Password

Password *

Re-Enter Password *

Terms and Conditions

☐ I have read the [Privacy Statement](#) and consent to this agreement. *

☐ I have read and understand the Terms and Conditions of Document Center. *

*Required

[Register](#)

powered by reCAPTCHA

Verify Your Telephone Number

You can change your telephone number. Please type: +[country code] followed by the area code without the leading zero, followed by the subscriber number. Example: +0000 XXX XXXXXXXX

The Document Center application requires telephone verification. We have sent a code to your telephone number. Please enter the code you have received and choose Continue.

Telephone *

Code *

*Required

[New Code](#) [Continue](#)

Figure 41: Protecting Self-Registration with Google reCAPTCHA / Phone Verification

The following access protection is provided for applications:

Protecting the registration to applications from spam and abuse

Preventing bots from automated fake user registrations to your applications

Google reCAPTCHA and phone verification provide further protection through self-registration

Logon Overlays in Customer Applications

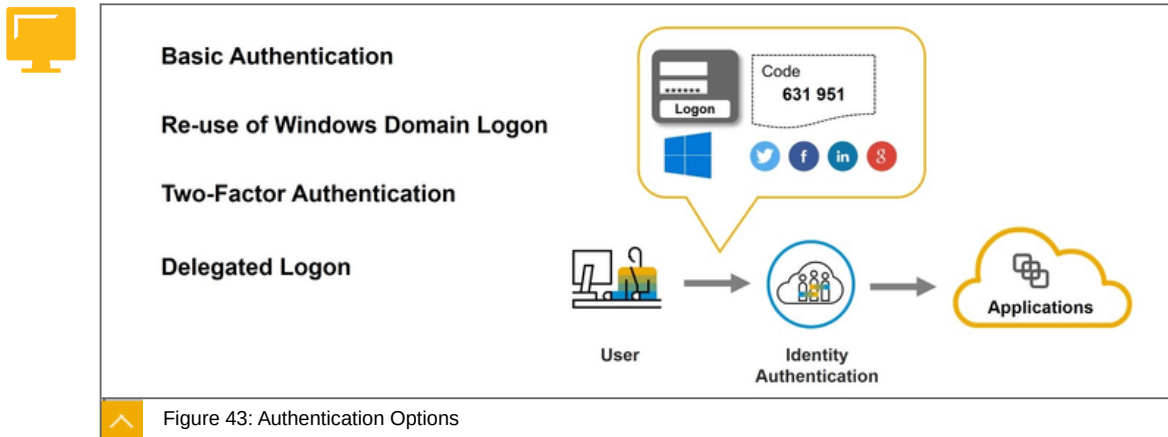


Logon Screen as an overlay (compared to a browser redirect to navigate away from application)

Figure 42: Logon Overlays in Customer Applications

The customer application logon screen can be programmed to integrate with the application. It has out-of-the-box integration with SAP Cloud portal.

Authentication Options



IAS provides the following authentication features:

Basic authentication

- Based on user ID/email and password

Re-use of Windows domain logon

- Kerberos token used for SSO

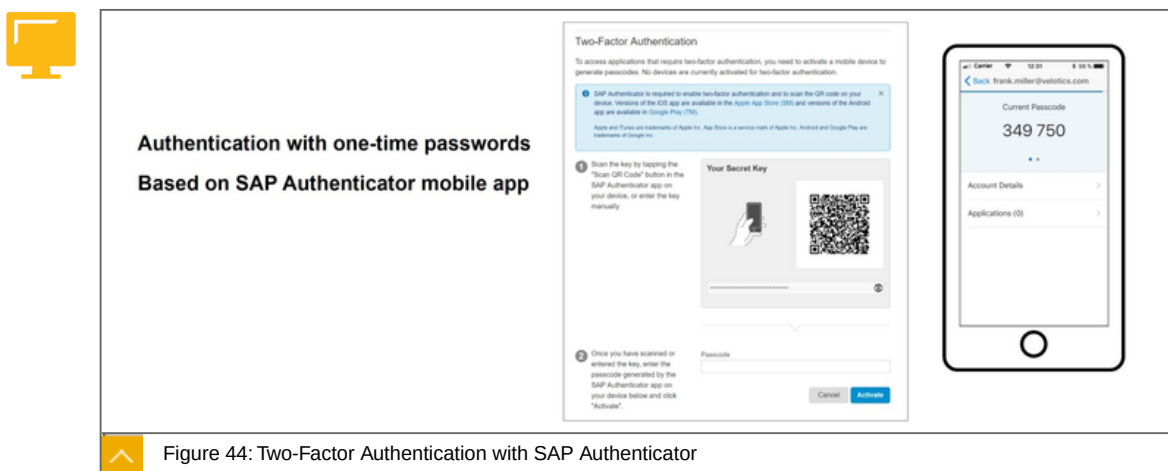
Two-factor authentication

- Second factor authenticated on mobile device

Delegated logon

- Social IdPs
- Corporate IdP

Two-Factor Authentication with SAP Authenticator



OTP is required for login in addition to the password or security token. The second factor authentication is used in high security scenarios.

The SAP Authenticator mobile app creates a One-time Password (6-digit) on a mobile device. It is available for iOS and Android and is RFC 6238 compatible (with authenticator apps from Google and Microsoft).

Control Access to the Application

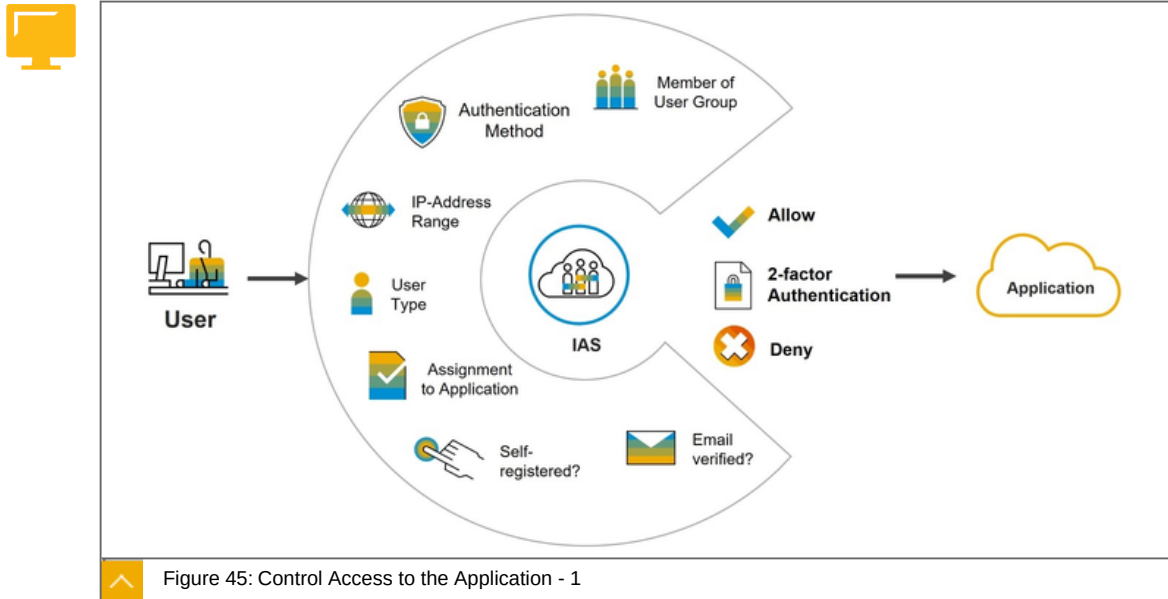


Figure 45: Control Access to the Application - 1

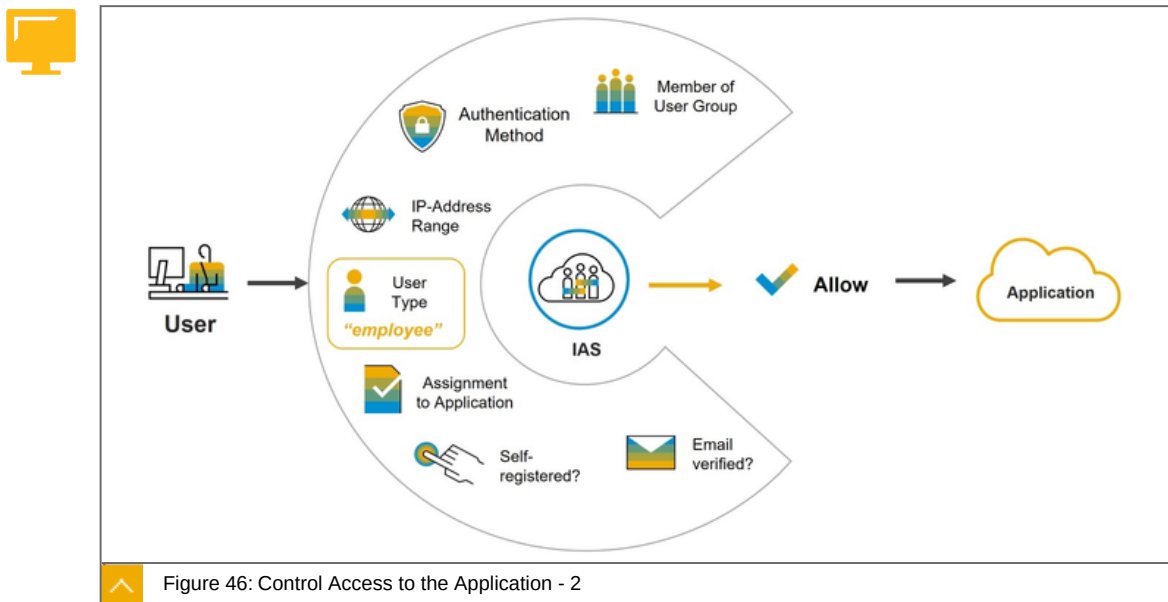
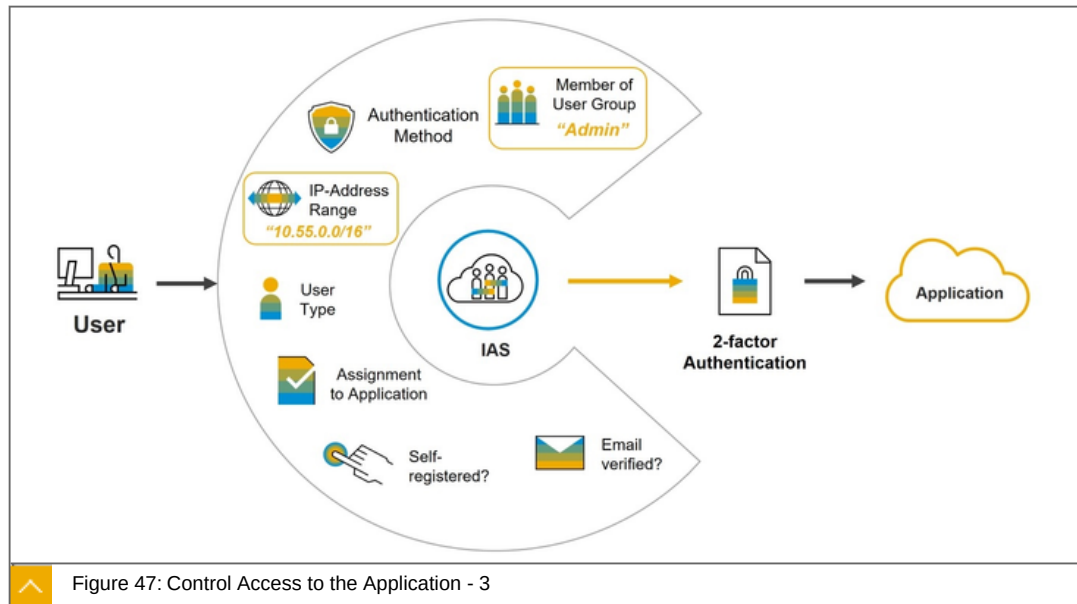


Figure 46: Control Access to the Application - 2



Custom Password Policy Configuration



Custom password policies can be configured

Custom Password Policy

Password Policy Name:

Password Length: Minimum: Maximum:

Password Lifetime: Minimum: Maximum:

Maximum Duration of User Inactivity:

Number of Last Used Passwords that Cannot Be Reused:

Number of Allowed Failed Login Attempts:

Password Locked Period:

Figure 48: Custom Password Policy Configuration

OAuth



OAuth is an **open standard** for access delegation.

It is used as a way for users to **grant web application access** to their information on another web application but **without giving them the passwords**.

“OAuth 2.0 focuses on client developer simplicity while providing **specific authorization flows** for web applications, desktop applications, mobile phones, and living room devices.”

OAuth 2.0 has introduced **scopes** to limit the access of an application to the account or data of an user.

Source: <https://oauth.net/2/>



Figure 49: OAuth

OAuth Authorization Flow

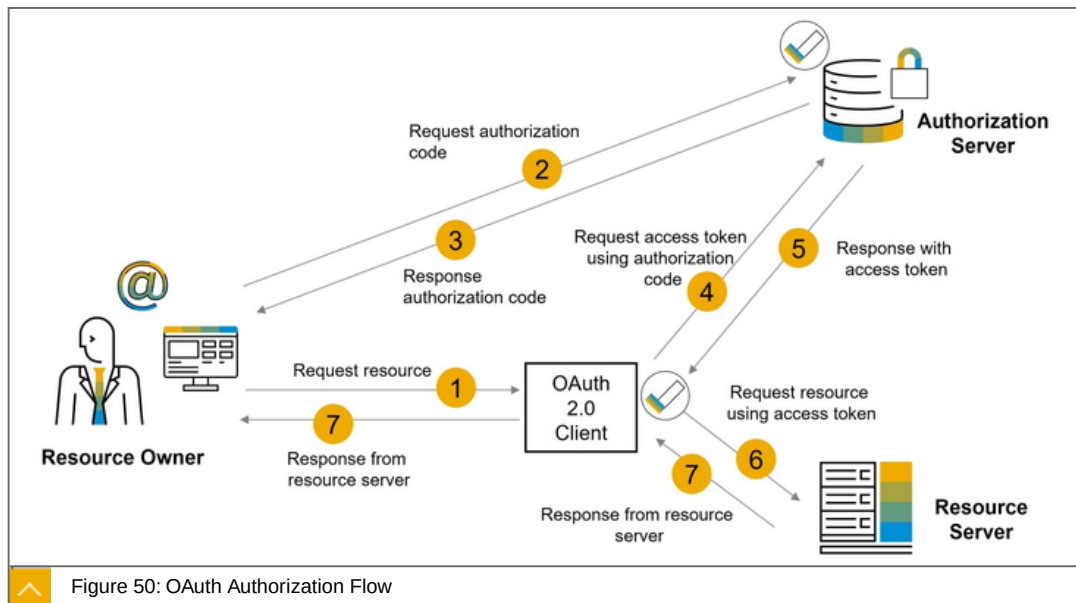


Figure 50: OAuth Authorization Flow

The OAuth authorization process is as follows:

1. The resource owner makes a call to protected resources using the OAuth 2.0 client.
2. The OAuth 2.0 client doesn't have an access token for the target system and redirects the browser to the authorization endpoint of the authorization server. This redirection is called authorization code request. At the authorization server, the resource owner is authenticated and can further restrict access or grant access to the preselected scopes.
3. The authorization server sends the authorization code back to the OAuth 2.0 client by redirecting the resource owner's user agent back to the redirection URI (that was defined during OAuth 2.0 client registration).

4. The OAuth 2.0 client sends an access token request to the authorization server's token endpoint. This access token request contains the authorization code.
5. The authorization server receives the access token request at its token endpoint and validates the authorization code. After a successful validation, the authorization server returns an access token to the OAuth 2.0 client.
6. The OAuth 2.0 client uses the access token to request resources.
7. The resource server grants the OAuth 2.0 client access to protected resources in accordance with the access token.



LESSON SUMMARY

You should now be able to:

Understand SAP Cloud Platform Identity Authentication and Usage Scenarios

Configure Branding and Customization

Configure Authentication Options

Learning Assessment

1. For which operation is IAS responsible for?

Choose the correct answers.

- ☐ **A** Authentication
- ☐ **B** Provisioning
- ☐ **C** SSO

2. Which of the following is NOT a valid way of creating users in IAS?

Choose the correct answer.

- ☐ **A** Self-registration
- ☐ **B** OData interfaces
- ☐ **C** SCIM

3. Which of the following protocols are available within IAS for authentication to third party applications/systems?

Choose the correct answers.

- ☐ **A** SCIM
- ☐ **B** SAML
- ☐ **C** OpenID Connect
- ☐ **D** REST

Learning Assessment - Answers

1. For which operation is IAS responsible for?

Choose the correct answers.

- ☒ **A** Authentication
- ☐ **B** Provisioning
- ☒ **C** SSO

This is correct! IAS is responsible for authentication and SSO operations.

2. Which of the following is NOT a valid way of creating users in IAS?

Choose the correct answer.

- ☐ **A** Self-registration
- ☒ **B** OData interfaces
- ☐ **C** SCIM

This is correct! OData interfaces is NOT a valid way of creating users in IAS.

3. Which of the following protocols are available within IAS for authentication to third party applications/systems?

Choose the correct answers.

- ☐ **A** SCIM
- ☒ **B** SAML
- ☒ **C** OpenID Connect
- ☐ **D** REST

This is correct! SAML and OpenID Connect are the protocols within IAS for authentication to third party applications/systems.

UNIT 5

S/4HANA On-Premise

Lesson 1

Understanding User Management in S/4HANA On-Premise

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UNIT OBJECTIVES

Understand S/4HANA On-Premise

Unit 5

Lesson 1

Understanding User Management in S/4HANA On-Premise



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Understand S/4HANA On-Premise

User Management in S/4HANA On-Premise

SAP S/4HANA's code line, simplified data model, and modern UX are consistent for both Cloud and on-premise. Designed for in-memory, SAP S/4HANA brings new business capability while simplifying the IT landscape.

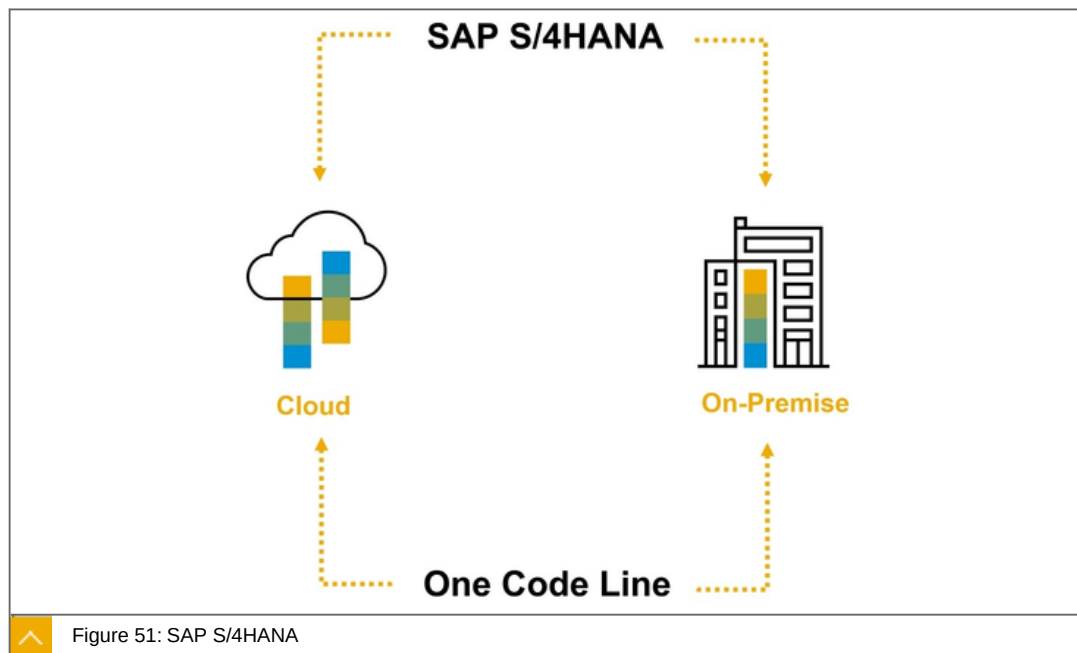


Figure 51: SAP S/4HANA

Items to consider when selecting an SAP S/4HANA solution:

- Business functionality
- Regulatory, industry, and regional requirements
- Individualization options
- IT strategy
- Innovation cycles

Adoption/upgrade efforts

Total Cost of Ownership (TCO)

Commercial models

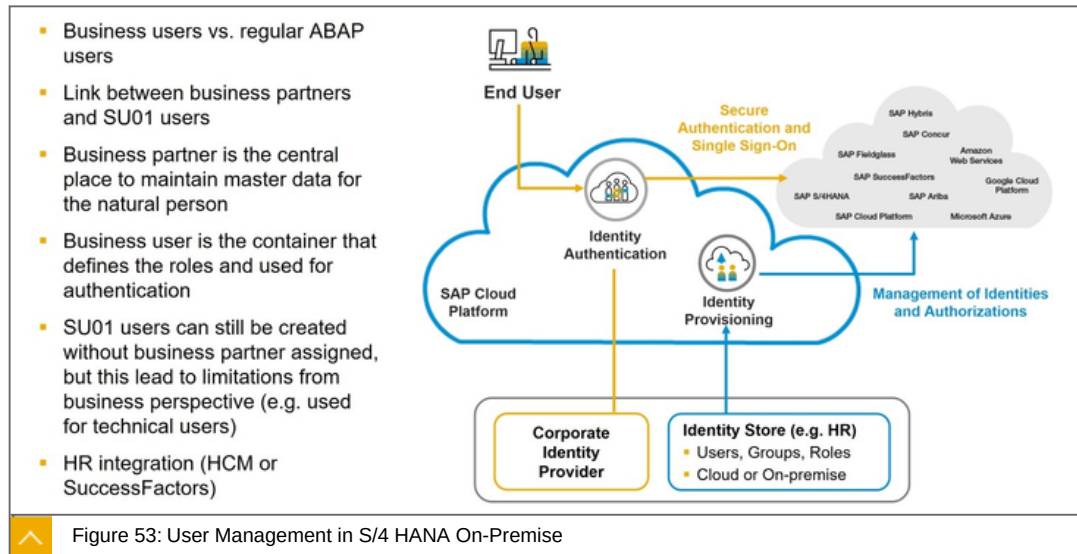
Deployment Options



	 SAP S/4HANA Cloud	 SAP S/4HANA Cloud, single-tenant edition	 SAP S/4HANA
Business Process	Standardized, core ERP	Flexible, ext. ERP	Customizable, ext. ERP
Innovation Lifecycle	Quarterly	Semi-annual	Annual, customer-led
TCO	Lowest	Lower	Higher
System governance	SAP-led	Customer-influenced	Customer-led
IT Infrastructure	SAP, Public	SAP, dedicated	Customer-managed
Customization	Within Standards	Within Standards	Open to Modifications
Extension	PaaS, SCP	PaaS, SCP	Open, SCP
System delivery	New implementation	New implementation	New or ECC conversion

Figure 52: Deployment Options

User Management in S/4 HANA On-Premise



SAP S/4HANA is introducing a new identity model for business users. A business user is defined as a natural person who is represented by a business partner and a link to a user in the system. Business users interact with the software in the context of a business process, for example, in the role of a purchaser, a sales representative, or a production planner.

Following the Principle-of-One in SAP S/4HANA, the business partner is the contact that maintains the user data for the person, work center, and communication-related information.

SAP S/4HANA Business User Management enables and supports the entire lifecycle maintenance of business users with integration to the employee lifecycle in the sense of organizational changes, leaving, or retirement. A user in SAP S/4HANA is now related (1:1) to a corresponding business partner (person) from whom essential data is referenced. This reduces redundant maintenance and prevents the disadvantage you have with outdated information.

The technical user is a SU01 user in the system, but also has a 1:1 relation to the corresponding business partner. The business user concept is used in many new applications in SAP S/4HANA. SU01 users created without business partners will lead to limitations because the business user is a prerequisite for many business applications and Fiori apps.

This business partner contains the personal information. The decoupling of user authentication and personal information has the advantage that it integrates the entire lifecycle of that person without redundant maintenance of the user data. Users can still be managed using the SU01 transaction or an Identity Provider (IdP) system replicated to all systems.

Information about people, work centers, and communication is derived from the related business partner (1:1 relation). In contrast to the Business Suite, this data is no longer redundantly managed within the User Management SU01 transaction, and so on.

In the Business Suite, you can use the BP (Business Partner Maintenance) transaction to assign users to business partners. In SAP S/4HANA this is no longer possible, so you avoid an inconsistent data model for business users.

With Business Partner Integration, the following changes to business user data in SAP S/4HANA are integrated and can no longer be edited in SU01:

Person

The personal data for the business user in SU01 is derived from the corresponding business partner, integrated with data that comes from the related employee from the HR system (SAP SuccessFactors or SAP HCM). This data is mapped from the HR system to the business partner.

Work Center

The work center data for the business user in SU01 is derived from the workplace address of the corresponding business partner. This address information is integrated with the related employee (for the Function and Department fields.) This data is replicated from the HR system to the business partner.

Communication

The communication data for the business user in SU01 is derived from the workplace address of the corresponding business partner. This communication information is integrated with the related employee information (for the Phone and E-Mail Address fields). This data is replicated from the HR system to the business partner.

Company

The organization data for ORGANIZATION_NAME and LOCATION for the business user in SU01 is derived from the company address of the corresponding business partner. In case of conversion from SAP ERP to SAP S/4HANA, this data is copied from the formerly assigned company address of the user via the periodically scheduled sync report /SHCM/RH_SYNC_BUPA_FROM_EMPL.

The benefit of automated integration is to get rid of the manual process that has the risk of causing inconsistencies in the system. Therefore, you can no longer maintain the following fields in the BP transaction:

Personnel Number (BUS000EMPL-EMPLOYEE_ID)

User (BUS000EMPL-USERNAME), for example, shown for role category WFM001 'Resource'

Internet User (BUS000EMPL-INTERNETUSER), for example, shown for role category BUP005 'Internet User'

The following role categories cannot be assigned to business partners in the BP transaction:

Employee (BUP003)

Freelancer (BBP010) Contingent worker

Service Performer (BBP005) Contingent worker with additional supplier relation (staffed from external supplier)

The above business partners roles cannot be assigned as they have special behavior regarding authorization management, which is driven by GDPR/DPP requirements. There is no replication or distribution (no replication from S/4 to any system).

For business partners which are assigned to one of these role categories, the fields of the following field groups cannot be maintained via the BP transaction or the customer/supplier master Fiori UIs (independent from whether HR integration is active or inactive):

Field group 1: Partner Number (+ Role for Change Doc. Display)

Field group 8: Form of Address

Field group 9: Bank Details (for bank detail's ID HC*)

Field group 25: Person: First Name

Field group 26: Person: Academic Title

Field group 29: Person: Correspondence Language

Field group 30: Person: Complete Name Line

Field group 33: Person: Gender

Field group 38: Person: Birth Name

Field group 41: Person: Middle Name

Field group 43: Person: Initials of Person

Field group 54: Person: Last Name

Field group 61: Address: Address Usage (HCM001, HCM002); note that the address data can be edited in transaction BP but will be overwritten by the periodically scheduled sync report /SHCM/RH_SYNC_BUPA_FROM_EMPL.

Relationships of the BUR025 relationship category, Service Provider, also cannot be maintained for roles of the BBP005 role category, Service Performer.

Business partners which are assigned to one of these role categories cannot be maintained using SAP MDG for Business Partners, Customers or Suppliers. The usage of such Workforce Business Partners in MDG is also restricted by the following conditions

In partner function maintenance: Only the ID of the Workforce Business Partners is visible, and navigation to that BP is not possible.

In BP relationships: Relationships to Workforce Business Partners are not visible and can't be maintained.

HR Integration is Active: Creating Business Partners (Persons) Related to Business Users.

Prerequisite: The HR integration switch in SAP S/4HANA is activated with the T77S0 system table and the configuration switches HRALX/HRAC and HRALX/PBPON. The HR system is the origin of business users.

If there is an active HR integration, the employees and contingent workers are replicated to SAP S/4HANA (from SAP SuccessFactors Employee Central or SAP Human Capital Management (HCM) on-premise) and the person's information is integrated into the corresponding business partner in SAP S/4HANA.

The synchronization of employee and employment information between HR and SAP S/4HANA is managed with the periodically scheduled sync report /SHCM/RH_SYNC_BUPA_FROM_EMPL. This report synchronizes employee and employment data into the business partner and user management consistency and provides a consistent data model.

Depending on the purpose of the person, the business partner is created in a role of one of the following role categories:

Employee (BUP003)

Freelancer (BBP010)

Service Performer (BBP005)

In addition, the supplier information is also created for payment-related processes. These roles cannot be assigned to existing business partners in the BP transaction.

If there are multiple employment conditions (concurrent / global employment / country transfer), the following must be noted:

If an employee (PERNR) has a user assigned (via infotype 105, subtype 0001), after the sync report, the user will be linked with the business partner which is linked to the central person and the central person is linked to PERNR.

One employee should have one user assigned (even in case of multiple assignments in case of concurrent / global employment / country transfer).

In case of a multiple assignment scenario and if an employee has different users assigned for different PERNR's, then a sync report will return an error. Such errors are reported in the application log. The log is not displayed automatically on employee replication in Cloud or employee synchronization report execution in onPremise.

The onPremise application log can be accessed via the SLG1 (or SLGD) transaction for the SHCM_EE_INTEGRATION object; subobject: BUPA_SYNC (filter via External ID field to Personnel Number-PERNR).

Human resources integration is inactive; creating business partners (persons) related to business users is inactive. The HR integration switch in SAP S/4HANA is de-activated via the T77S0 system table with the HRALX/HRAC and HRALX/PBPON configuration switches.

Business users and persons can be created in SAP S/4HANA (as of release 1709 FPS01) using the Maintain Employees Fiori app. With this app you can create and update business partners and their assignments to users to create and manage business users. The app provides a mass maintenance via CSV upload.

For mass maintenance, you can also use the report RFS4BPU_EMPLOYEE_UPLOAD with CSV upload for a mass import (SAP S/4HANA1610). With SAP S/4HANA1809 you can use the RFS4BPU_IMPORT_BUSINESS_USER report.

If HR integration is inactive in SAP S/4HANA, the user data can still be managed with the SU01 transaction.



LESSON SUMMARY

You should now be able to:

- Understand S/4HANA On-Premise

Learning Assessment

1. Which connector in IPS should be used if you would like to create and maintain business partners in SAP S/4HANA on-premise?

Choose the correct answer.

- ☐ A ABAP connector
- ☐ B S/4HANA on-premise
- ☐ C SuccessFactors

2. What is the cardinality of the relation between business partner and SU01 user?

Choose the correct answer.

- ☐ A 1:n
- ☐ B n:m
- ☐ C 1:1

3. Which of the following systems are supported for HR integration with S/4HANA?

Choose the correct answers.

- ☐ A S/4HANA Cloud
- ☐ B SuccessFactors
- ☐ C HCM
- ☐ D Identity Management

Learning Assessment - Answers

1. Which connector in IPS should be used if you would like to create and maintain business partners in SAP S/4HANA on-premise?

Choose the correct answer.

- ☐ A ABAP connector
- ☒ B S/4HANA on-premise
- ☐ C SuccessFactors

This is correct! SAP S/4HANA on-premise should be used if you would like to create and maintain business partners in SAP S/4HANA on-premise.

2. What is the cardinality of the relation between business partner and SU01 user?

Choose the correct answer.

- ☐ A 1:n
- ☐ B n:m
- ☒ C 1:1

This is correct! 1:1 is the cardinality between business partner and SU01 user.

3. Which of the following systems are supported for HR integration with S/4HANA?

Choose the correct answers.

- ☐ A S/4HANA Cloud
- ☒ B SuccessFactors
- ☒ C HCM
- ☐ D Identity Management

This is correct! SuccessFactors and HCM are supported for HR integration with S/4HANA.